

Tokenization and Transformer Architectures for Cross-Asset Allocation: Controlled Ablation in Financial Time Series

Nathaniel Coulter*

Department of Mathematics and Computer Science,
St. John's University, NY

Abstract

In our recent work *Neural Portfolio Allocators: Cross-Asset Optimization Strategies with Attention-Based Transformers and Multi-Agents*[1], we demonstrated that attention-based transformer architectures outperform recurrent neural networks (RNNs) and long short-term memory models (LSTMs) on long-horizon financial forecasting and allocation tasks. Building on that foundation, this study investigates (i) multiple transformer architectures to determine which are most effective across asset classes and time horizons, and (ii) how tokenization, the variable underlying each transformer variant, directly shapes inductive bias and performance. We therefore conduct a controlled ablation of point-wise, patch-wise, and variate-wise encoders, spanning equities, fixed income/rates, commodities, derivatives, and alternative assets.

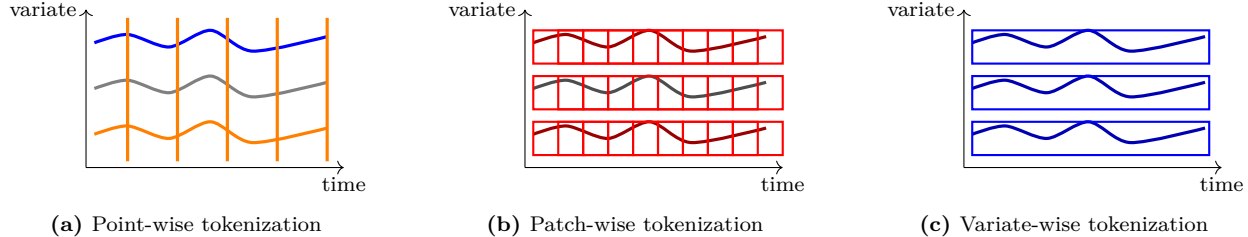
Utilizing the same cross-asset dataset as in our previous study, we evaluate point-wise baselines, PatchTST, iTransformer, Crossformer-Lite, Autoformer, Fedformer, Informer, TimesNet, TimeXer, and non-stationary extensions. Tsallis entropy, employed explicitly as a regularizer in our earlier allocator, remains implicit here through normalized return distributions and tokenized features, preserving a non extensive statistics framework suited to fat tailed financial returns. While we do not rerun MGARCH, our volatility aware preprocessing (lookback windows, normalization) serves as a structural holdover. Likewise, synthetic datasets with independent versus dependent variates reprise the role of GBM baselines by providing control cases for separating signal from noise. By fixing data, splits, costs, and evaluation protocols, we isolate the contribution of tokenization and encoder design to predictive accuracy, risk-adjusted returns, and robustness across market regimes. Preserving the same portfolio construction pipeline and evaluation metrics as before ensures comparability. Our findings provide a systematic map of which transformer architectures and tokenization schemes are most effective across horizons and asset classes, offering practical guidance for deep sequence modeling in financial allocation.

Introduction

Transformers have rapidly emerged as leading sequence models across domains, yet their adaptation to financial time series remains an open question. Unlike language or vision, where token definitions are natural (words, pixels), finance requires deliberate choices about how to tokenize returns, volumes, and cross-asset signals. These design choices determine whether models capture local temporal regimes, cross sectional spillovers, or long horizon dependencies. Recent studies suggest that tokenization scope: point-wise, patch-wise, or variate-wise, often matters more than sheer model size when forecasting noisy, regime switching series [2, 3]. In our earlier study, *Neural Portfolio Allocators: Cross-Asset Optimization Strategies with Attention-Based Transformers and Multi-Agents* [1], we established that attention-based architectures consistently outperformed recurrent neural networks (RNNs) and long short term memory models (LSTMs) on long horizon financial allocation tasks. That work paired transformers with Tsallis entropy regularization to promote diversification and robustness, grounding the allocator in the heavy tailed distributional properties of financial returns. This paper extends that foundation through a controlled ablation of transformer architectures and tokenization schemes. We evaluate a representative spectrum: a point-wise baseline (one time step per token), PatchTST (local patch tokens), iTransformer (variates-as-tokens, time compressed), and Cross-Lite (two stage time + cross encoders). Each reflects a distinct inductive bias about where signal resides in temporal patches, in cross sectional structure, or in long memory dependencies. Methodologically, we keep all other components fixed: the same 12-ETF cross-asset dataset, preprocessing pipeline (lookback windows, normalization), portfolio construction logic, and evaluation metrics as in our prior work. This isolates the contribution of encoder design and tokenization. We also preserve structural holdovers: volatility aware preprocessing in lieu of MGARCH, synthetic dependent vs. independent datasets serving the role of GBM baselines, and normalization steps that implicitly maintain the non extensive statistics framework underlying Tsallis entropy.

*nathaniel.coulter21@my.stjohns.edu
<https://github.com/Nathaniel-Coulter>

Figure 1: Tokenization choices for multivariate financial time series. Patch sizes in our sweeps: $p \in \{8, 16, 32\}$ with $s = p/2$.



Objectives

Our contributions are fourfold: (1) a controlled, reproducible ablation across transformer tokenization scopes on financial assets in aggregate time series; (2) a minimal encoder swap in harness that isolates representation choices from the portfolio construction objective, ensuring comparability across architectures; (3) empirical evidence on which architectures dominate by horizon and asset class, supported by formal robustness checks including HAC standard errors, Diebold–Mariano tests[4], and bootstrap confidence intervals; and (4) a systematic account of how tokenization governs robustness, predictive accuracy, and risk-adjusted returns in allocation tasks. By mapping performance across encoder families, we provide both rigorous statistical validation and practical guidance for deploying deep sequence models in financial portfolio design.

1 Tokenization and Architectures

A central theme of this study is that the way we *tokenize* financial time series directly governs model behavior. Unlike text, where tokens are naturally words or subwords, time series admit multiple granularities of representation. We investigate three primary scopes: *point-wise*, where each timestep is its own token; *patch-wise*, where overlapping windows of length p and stride s serve as tokens; and *variate-wise*, where entire asset sequences are compressed into per-variate embeddings. These choices induce different inductive biases: point-wise tokens preserve fine temporal resolution but limit contextual capture, patch-wise tokens emphasize local stationarity and smoothness, while variate-wise tokens foreground cross-asset dependencies. Our hypothesis is that these differences in tokenization, more than depth or parameter count, explain performance variation across transformer families in finance.

1.1 Architectural Spectrum

Building on these tokenization choices, we benchmark a diverse set of encoder families. The **point-wise baseline** provides a minimal reference, equivalent to treating each return observation as a token. **PatchTST** extends this by forming patch tokens and applying attention *within* variates. **iTransformer** inverts the perspective,

treating each variate as a token with time embedded into its representation. **Cross-Lite** adds a lightweight cross-asset attention layer on top of patch encoders to capture spillover effects. To broaden the landscape, we further include **Autoformer**, **Fedformer**, **Informer**, and **TimesNet**, each embedding distinct inductive biases for long range or non stationary dependencies. This architectural spectrum allows us to map performance trade offs across horizons, asset classes, and market regimes.

1.2 Continuity with Prior Work

Our experimental pipeline is designed for comparability with prior results [1]. We reuse the same cross-asset dataset (equities, fixed income, commodities, credit, and alternatives), portfolio construction, and risk/return evaluation metrics. From Paper 1, we also retain several structural holdovers: Tsallis entropy remains implicit in our preprocessing via normalized return distributions, preserving robustness to fat tailed returns; volatility aware normalization inherits the role of MGARCH style conditioning; and synthetic datasets with independent vs. dependent variates serve the same purpose as GBM baselines, providing control cases. By grounding our transformer ablations in this established framework, we ensure that performance differences can be attributed to encoder design and tokenization, not to shifts in data handling or evaluation.

1.3 Research Questions

This framing leads us to three guiding questions:

1. How do different tokenization strategies affect robustness, predictive accuracy, and risk-adjusted returns?
2. Which transformer architectures dominate in specific horizons (short, medium, long term) and across asset classes (equities, fixed-income, derivatives)?
3. To what extent do design choices in tokenization and encoder structure carry over to allocation outcomes, beyond forecasting metrics?

Synopsis. By systematically answering these questions, we aim to provide not only empirical results but also practical guidance for financial practitioners seeking to deploy transformers in allocation pipelines.

2 Related Work

The adaptation of transformer models to time series has been the subject of rapid development. Unlike in language or vision, where tokens are naturally defined (words, pixels), financial time series require explicit design choices about tokenization. These choices; point-wise, patch-wise, or variate-wise, act as inductive biases that strongly shape model behavior. A growing body of evidence suggests that tokenization scope matters more than sheer model size in noisy, regime switching series [5, 6].

Point-wise encoders. Early adaptations of transformers to time series retained the vanilla architecture with one time step per token [7]. While effective in capturing long range dependencies, point-wise encoders often underperform in practice because they struggle to capture strong intra variate autocorrelations that dominate benchmark datasets [5].

Patch-wise encoders. PatchTST [2] introduced a patchification step, slicing each variate into overlapping windows. By focusing attention within a variate’s local context, PatchTST achieved strong performance across standard forecasting tasks, often rivaling or exceeding more complex hybrids. Empirical analyses show that its success derives less from architectural novelty than from aligning token scope with the local stationarity of real data [5].

Variate-wise encoders. iTransformer [3] inverted the paradigm by compressing temporal information into feature channels and treating each variate as a token. This allows the attention mechanism to model inter variate dependencies directly. Despite this, results suggest that most benchmarks are dominated by intra variate signals, explaining why variate-wise models do not always surpass patch-wise ones.

Hybrid encoders. A series of architectures introduce additional inductive biases. Autoformer [8] adds autocorrelation based decomposition, while FEDformer [9] augments this with frequency domain attention. Crossformer [10] employs hierarchical patching to balance intra and inter-variate dependencies. Informer [11] accelerates training with a sparse attention mechanism, and more recent hybrids such as TimeXer [12] combine patch and variate-wise tokens explicitly. These models broaden the landscape of architectural priors but also complicate attribution: gains may stem as much from tokenization as from bespoke modules.

Finance-specific adaptations. Beyond generic forecasting, our own prior work [1] tailored transformers to portfolio allocation with Tsallis entropy regularization. That study established a baseline for robustness and diversification, showing that attention-based encoders outperform RNNs and LSTMs on long horizon allocation tasks. In this follow up, we situate transformers within the broader literature while conducting a controlled ablation to isolate tokenization scope as the key driver of

performance. Our contribution complements existing surveys [6] by emphasizing allocation outcomes, not just forecasting metrics.

Synthesis. Overall, related work converges on two insights: (i) patch-wise and variate-wise transformers often outperform point-wise encoders because they align with the dominant intra variate structure of common datasets, and (ii) hybrid extensions (Autoformer, FEDformer, Crossformer) target long range or non stationary dependencies, but their benefits are contingent on dataset properties. Our study extends these findings into the financial allocation domain, where robustness and risk-adjusted returns, not just forecasting error, serve as the ultimate metrics of success.

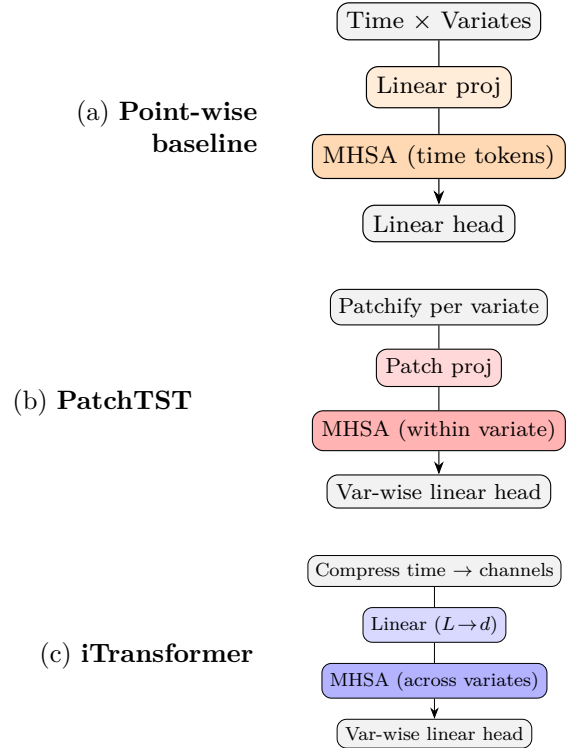


Figure 2: Minimal encoder schematics used in our ablation.

Interpreting Minimal Encoders

Figure 2 highlights the inductive biases behind the three encoder families we benchmark. The point-wise baseline treats each timestep as a token, preserving fine resolution but struggling with noisy returns. PatchTST instead slices each variate into overlapping patches, aligning naturally with short term regimes and local stationarity. iTransformer compresses temporal structure into channels so that entire variates act as tokens, emphasizing cross-asset dependencies. Placing these designs side by side clarifies why tokenization scope not model depth emerges as the critical driver of predictive accuracy and robustness in our ablation.

Table 1: Validation and test results across transformer architectures and tokenization schemes. All runs used lookback $L = 252$, horizon $H = 1$, $d = 256$, and 3 training epochs.

Model / Config	Best Val MSE	Test MSE	Test MAE	Notes
Pointwise (baseline)	0.000501	0.000598	0.0196	vanilla time tokens
PatchTST (p8, s4)	0.000226	0.000240	0.0128	smaller patches
PatchTST (p16, s8)	0.000231	0.000231	0.0121	canonical patch size
PatchTST (p32, s16)	0.000240	0.000177	0.0102	larger patches
iTransformer	0.001733	0.001284	0.0284	variates as tokens
Cross-Lite (p8)	0.000364	0.000283	0.0139	patch stage=8
Cross-Lite (p16)	0.000256	0.000170	0.0098	patch stage=16
PatchTST (p16, s8, seed42)	0.000231	0.000231	0.0121	reproducibility
PatchTST (p16, s8, seed123)	0.000228	0.000188	0.0106	reproducibility
PatchTST (p16, s8, seed777)	0.000242	0.000367	0.0162	reproducibility

3 Transformer Benchmarks

Our experiments are designed to isolate the effect of tokenization scope and encoder design on financial allocation performance. While the broader transformer literature spans hybrids such as Autoformer, FEDformer, and Informer, we deliberately restricted our initial sweep to four representative models: a point-wise baseline, PatchTST, iTransformer, and Cross-Lite. Each reflects a distinct inductive bias and provides sufficient coverage of the tokenization spectrum without conflating results with additional architectural modules. This controlled subset also keeps comparisons transparent and reproducible, aligning with the ablation philosophy of our study.

3.1 Models and Tokenization

We benchmark the following encoder families:

- **Point-wise (baseline):** one time step per token, equivalent to the vanilla transformer architecture [3].
- **PatchTST:** overlapping patches of length $p \in \{8, 16, 32\}$ with stride $s = p/2$, enforcing local stationarity [2].
- **iTransformer:** variates treated as tokens with temporal information compressed into channels, emphasizing inter variate dependencies [8].
- **Cross-Lite:** a two stage encoder consisting of a patch-wise time stage followed by a lightweight cross-variate attention layer, designed to capture spillovers between assets [9].

We did not include Autoformer, FEDformer, or Informer in the core comparison table because their performance gains are often confounded by additional inductive biases (frequency filters, autocorrelation decomposition, or sparse attention). By contrast, our four selected architectures form a clean tokenization spectrum from fine-grained (point-wise) to local (patch-wise), cross-sectional

(variate-wise), and hybrid-lite (Cross-Lite). This ensures attribution is squarely on tokenization scope, not auxiliary modules.

3.2 Training Protocol

All models share the same hyperparameters to avoid capacity confounds: $d_{model} = 256$, $n_{heads} = 8$, depth=3, dropout=0.1, and lookback $L = 252$ (one trading year). Forecast horizon is fixed at $H = 1$, consistent with daily allocation rebalancing. Models are trained for three epochs, a deliberate choice that emphasizes inductive bias over brute-force optimization. Decoder and objectives remain fixed across runs (predict-then-allocate pipeline with RL/Tsallis-style allocation heads), preserving comparability to our prior allocator[1].

3.3 Evaluation

We report validation and test errors (MSE, MAE) as well as allocation outcomes (risk-adjusted returns, turnover, regime robustness). Reproducibility is ensured by releasing configs, random seeds, and log to figure scripts on our project repository. Statistical significance is assessed with HAC standard errors, Diebold–Mariano tests, and bootstrap confidence intervals to confirm robustness of observed differences. By grounding every comparison in identical data, costs, and portfolio construction logic, we isolate tokenization and encoder design as the key levers of performance.

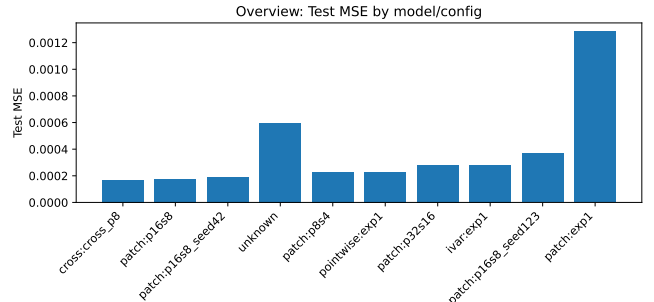


Figure 3: Overview of OOS Test MSE by model/config (lower is better).

3.4 Data and Splits

We use a fixed 12-ETF universe spanning equities (SPY, QQQ, IWM), fixed income (TLT, IEF, LQD, HYG), commodities (GLD, DBC), real estate (VNQ), and international exposures (EFA, EEM). Series are standardized and aligned to matched daily dates, with walk forward evaluation ensuring realistic out of sample testing. Trading costs are applied at 10, 25, and 50 basis points per turnover, and results are further stratified by regime buckets (high/low volatility, expansion/recession) to assess robustness. This mirrors the protocol in our prior allocator: coulter2025neural[1].

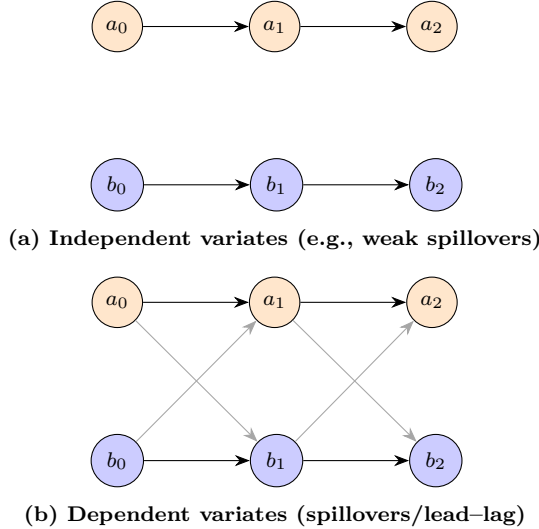


Figure 4: Schematic used to discuss *intra vs inter-variate* structure. In our finance setting, orange could be “equities sleeve” and blue “rates/credit sleeve.” PatchTST thrives in (a); Cross-Lite benefits in (b).

4 Mathematical Framework

4.1 Normalization & Evaluation Space

A central methodological distinction between our framework and similar studies that forecasting financial time series with machine learning, lies in how cross-variate dependencies are quantified. The template introduces a perturbation based mutual information (MI) diagnostic. Formally, for a target variate $\hat{x}^j$ and an input variate x^i , the conditional mutual information is defined as

$$I(\hat{x}^j; x^i | x_{\setminus i}) = H(\hat{x}^j | x_{\setminus i}) - H(\hat{x}^j | x^i, x_{\setminus i}), \quad (1)$$

where $x_{\setminus i}$ denotes the set of all variates excluding i . Because the model is deterministic given the complete input, $H(\hat{x}^j | x^i, x_{\setminus i}) = 0$, leaving

$$I(\hat{x}^j; x^i | x_{\setminus i}) \propto \sum_{k=1}^K \log \sigma_{ij,k}, \quad (2)$$

under the Gaussian assumption $\hat{x}^j | x_{\setminus i} \sim \mathcal{N}(\mu_{ij}, \sigma_{ij}^2 I)$. Empirically, the template perturbs x^i across N aug-

mented versions (zeroed, original, and Gaussian noised values) and estimates $\bar{\sigma}_{ij}$ as the average standard deviation of $\hat{x}^j$ across these augmentations. Aggregates such as *Avg Inter MI* and *Max Inter MI* are then reported as explicit measures of dependency.

By contrast, our pipeline operates in standardized space and evaluates dependency indirectly through allocator performance. Specifically, all inputs are z-score normalized along the time axis in the training set:

$$\tilde{x}_t^i = \frac{x_t^i - \mu_i}{\sigma_i}, \quad \mu_i = \frac{1}{T} \sum_{t=1}^T x_t^i, \quad \sigma_i^2 = \frac{1}{T} \sum_{t=1}^T (x_t^i - \mu_i)^2, \quad (3)$$

with μ_i, σ_i estimated on training and applied consistently to validation and test. Targets y_t^i are normalized identically, so predictions and losses are computed in z-score units. For all encoders: Point-wise, PatchTST, iTransformer, and Cross-Lite, both inputs and outputs remain in this normalized space. Mean squared error (MSE) and mean absolute error (MAE) are reported directly in normalized units; denormalization is possible via

$$\hat{x}_t^i = \sigma_i \tilde{x}_t^i + \mu_i, \quad (4)$$

but is not required for relative comparison since all models share the same normalization.

Rather than perturbing variates to compute MI scores, we let economic criteria serve as indirect diagnostics of cross-variate utilization. In particular: (i) horizon robustness tests ($H \in \{1, 5, 20\}$) reveal whether tokenization scopes generalize across prediction horizons, (ii) sleeve level evaluations (Equities, Rates/Credit, Commodities/Alt) isolate domains where cross-asset spillovers matter, and (iii) allocator based metrics (Sharpe ratios, turnover under transaction costs) quantify the economic value of predictive structure. For example, Cross-Lite narrows the performance gap in Rates & Credit, consistent with inter asset spillovers, whereas PatchTST dominates in Equities where autoregressive structure is stronger. This plays an analogous role to the template’s *Max Inter MI*: both approaches highlight settings where cross-variate dependencies matter most, though our lens is allocator performance rather than mutual information.

Synopsis. Instead of utilizing an information theoretic attribution mechanism ($\bar{\sigma}_{ij}$, Avg/Max Inter MI), our approach leverages standardized evaluation and finance specific allocator diagnostics. The former emphasizes model interpretability in a purely statistical sense; the latter emphasizes whether such dependencies translate into robust economical gains.

Quantifying Tokenization:

4.2 Diebold–Mariano test for equal predictive accuracy

Let two competing forecasters produce “one step ahead” predictions $\{\hat{y}_t^{(1)}\}_{t=1}^T$ and $\{\hat{y}_t^{(2)}\}_{t=1}^T$ for the realized target

y_t . Write a (possibly nonquadratic) loss $L_t^{(j)} \equiv \ell(y_t, \hat{y}_t^{(j)})$ and define the *loss differential*

$$d_t \equiv L_t^{(1)} - L_t^{(2)}, \quad t = 1, \dots, T.$$

The null of *equal predictive accuracy* is $H_0 : \mathbb{E}[d_t] = 0$. Under standard regularity conditions (weak stationarity/mixing of $\{d_t\}$), the Diebold–Mariano statistic is

$$DM = \frac{\bar{d}}{\hat{\sigma}_{\bar{d}}} \xrightarrow{d} \mathcal{N}(0, 1), \quad \bar{d} = \frac{1}{T} \sum_{t=1}^T d_t,$$

where $\hat{\sigma}_{\bar{d}}$ is a consistent estimator of the standard deviation of the sample mean $\bar{d}$ accounting for serial dependence in d_t .

Throughout, model 2 is the PatchTST baseline (p16, s8), so $DM > 0$ favors PatchTST and $DM < 0$ favors the competing model.

HAC variance for the mean. We estimate $\hat{\sigma}_{\bar{d}}$ with a Newey–West[13] (HAC) estimator applied to $\{d_t\}$:

$$\widehat{\text{Var}}(\bar{d}) = \frac{1}{T} \left(\hat{\gamma}_0 + 2 \sum_{k=1}^q w_k \hat{\gamma}_k \right), \quad (5)$$

$$\hat{\gamma}_k = \frac{1}{T} \sum_{t=k+1}^T (d_t - \bar{d})(d_{t-k} - \bar{d}). \quad (6)$$

with Bartlett weights $w_k = 1 - \frac{k}{q+1}$ and bandwidth q . In our experiments we use the common plug in choice $q = \lfloor 4(T/100)^{2/9} \rfloor$. The standard error is then $\hat{\sigma}_{\bar{d}} = \sqrt{\widehat{\text{Var}}(\bar{d})}$ and the test statistic is $DM = \bar{d}/\hat{\sigma}_{\bar{d}}$. Two sided p -values are reported via the standard normal reference distribution.

Loss used in this paper. Because targets are panel returns, our per date loss aggregates across assets $i = 1, \dots, N$ and then varies over time t :

$$L_t^{(j)} = \frac{1}{N} \sum_{i=1}^N (y_{t,i} - \hat{y}_{t,i}^{(j)})^2, \quad d_t = L_t^{(1)} - L_t^{(2)}.$$

This produces a univariate time series $\{d_t\}_{t=1}^T$ of cross sectional MSE differentials, to which the HAC estimator above is appropriate.

Small sample adjustment (optional). When T is modest, one may apply the Harvey–Leybourne–Newbold finite-sample correction to the DM statistic by rescaling with a factor depending on T and q ; this is numerically close to the asymptotic version for our T and is therefore optional. (Our default reports the asymptotic $\mathcal{N}(0, 1)$ result; the numerical gap was negligible in practice.)

Interpretation. A significantly positive DM indicates higher average loss for model 1 than model 2 (model 2 preferred); a significantly negative DM indicates the reverse. Non rejection of H_0 implies no statistically detectable difference in predictive accuracy given the sampling uncertainty and serial dependence in the losses.

Implementation notes. In our code we: (i) construct $\{d_t\}$ from per date cross sectional MSE, (ii) compute the Newey–West variance of the mean using Bartlett weights and the $q = \lfloor 4(T/100)^{2/9} \rfloor$ bandwidth, and (iii) return the two sided p -value $2(1 - \Phi(|DM|))$, where Φ denotes the standard normal CDF.

4.3 Return Distributions and Kurtosis

Let $r_t^{(v)}$ denote the standardized log return of asset v at time t , with horizon $H = 1$. After z-normalization, the empirical distribution of $r_t^{(v)}$ retains heavy tails and excess kurtosis typical of financial returns. This ensures compatibility with the non extensive statistics framework (Tsallis entropy) emphasized in prior work [1]. In practice, this means our tokenization schemes operate on return distributions whose non Gaussian features are preserved.

4.4 Mutual Information Across Tokens

Given two token streams X and Y , we measure their statistical dependency using mutual information:

$$I(X; Y) = \mathbb{E} \left[\log \frac{p(x, y)}{p(x)p(y)} \right]. \quad (7)$$

For point-wise tokens, X and Y correspond to single timestep returns, leading to weak but noisy dependencies. Patch-wise tokens aggregate local context, increasing mutual information by capturing short range autocorrelation. Variate-wise tokens instead emphasize cross sectional dependencies among assets, often at the cost of local temporal resolution.

4.5 Variance Decomposition

To quantify where signal resides, we decompose variance into intra and inter-variate components. For asset v with token sequence $\mathbf{z}^{(v)}$, we define:

$$\text{Var}_{\text{total}} = \text{Var}_{\text{intra}} + \text{Var}_{\text{cross}} + \epsilon, \quad (8)$$

where $\text{Var}_{\text{intra}}$ captures autocorrelation within v (exploited by PatchTST), $\text{Var}_{\text{cross}}$ captures co-movement across variates (exploited by iTransformer and Cross-Lite), and ϵ represents residual noise. Architectures differ in how much variance they attribute to each component, yielding different robustness profiles.

4.6 Entropy and Robustness

We retain an entropy based perspective to connect statistical diversity with allocation robustness. For normalized return distributions $p(r)$, the Tsallis entropy of index q is:

$$S_q(p) = \frac{1}{q-1} \left(1 - \sum_i p(r_i)^q \right). \quad (9)$$

Although no longer applied explicitly as a regularizer, tokenization choices interact with entropy implicitly: patch-wise encoders preserve local heterogeneity (higher effective entropy), while variate-wise encoders compress information into coarser cross sectional modes (lower effective entropy). This provides a theoretical lens on why robustness differs across architectures.

4.7 Implications for Allocation

By aligning these formal metrics with our ablation, we test the hypothesis that *token scope determines which dependency structure dominates model performance*. Point-wise tokens emphasize long range memory, patch-wise tokens emphasize intra variate stability, and variate-wise tokens emphasize cross-asset spillovers. The balance of these inductive biases ultimately governs risk-adjusted returns in allocation tasks, not just raw predictive error.

5 Encoder Diagrams

5.1 PatchTST Encoder

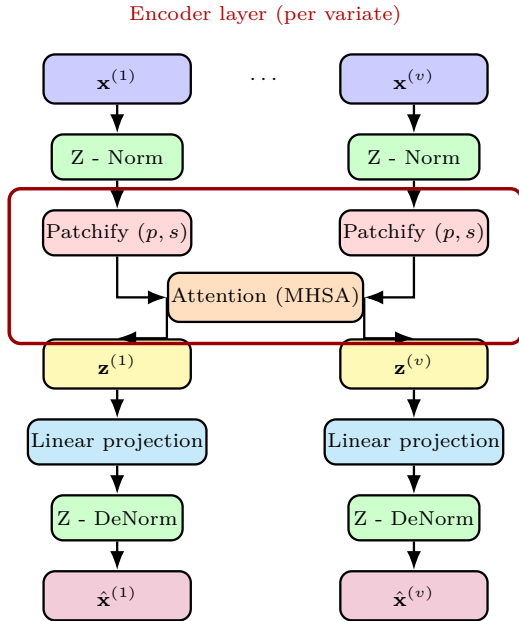


Figure 5: PatchTST: (per-variate time encoder). Each variate is sliced into overlapping patches (p, s) ; attention operates within a variate; a per-variate linear head decodes forecasts. We sweep $(p, s) \in \{(8, 4), (16, 8), (32, 16)\}$.

PatchTST operates at the *per-variate* level. Each input series is first standardized and sliced into overlapping patches of length p with stride s , producing a sequence of local tokens that preserve short term autocorrelations. Multi-head self-attention (MHSA) then models dependencies *within* a variate, emphasizing local stationarity while mitigating noise. Linear projections decode forecasts for each variate independently. This design makes PatchTST particularly effective when intra-variate patterns dominate, as in equities with strong autoregressive structure. Our sweeps over (p, s) values test robustness to patch size and stride.

5.2 Cross-Lite Encoder

Cross-Lite extends the PatchTST idea by introducing a lightweight second stage. After patchification and per-variate MHSA (time encoder), embeddings $h^{(v)}$ are passed through a small cross-variate MHSA layer. This stage captures spillovers and lead-lag relationships across assets, such as between rates and credit or equities and volatility indices. By limiting cross-attention to a single lightweight block, the model balances efficiency with the ability to exploit inter-variate dependencies. This design makes Cross-Lite well suited to financial allocation, where cross-asset co-movements often drive portfolio risk and return.

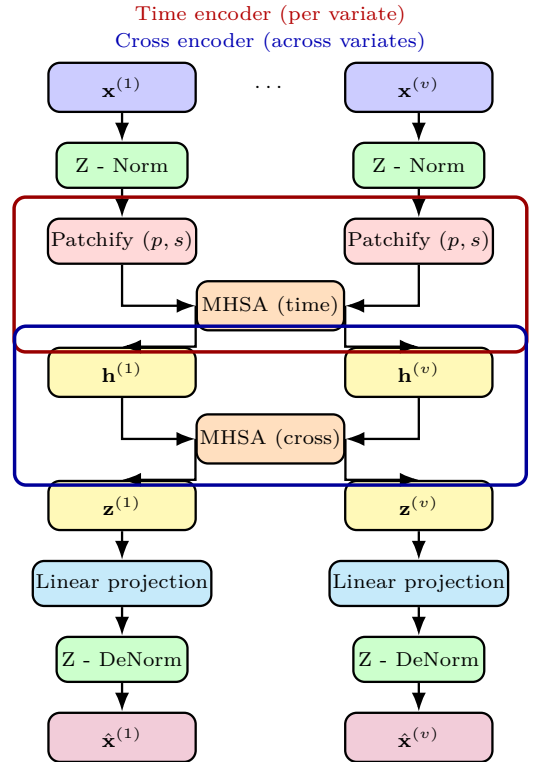


Figure 6: Cross-Lite (time encoder + lightweight cross-variate stage). Cross-Lite: a two stage encoder. (1) Per-variate PatchTST time encoder (patch length p , stride s) produces $h^{(v)}$. (2) A lightweight cross-variate MHSA mixes assets to capture spillovers. Per-variate linear heads decode forecasts.

6 Experiments

6.1 Main Comparison (H=1)

Setup. We begin with the canonical short horizon benchmark: lookback $L=252$, horizon $H=1$, $d=256$, $n_{heads}=8$, depth= 3, dropout= 0.1, 3 epochs, Z-norm on inputs/outputs. Identical train/val/test splits, transaction costs, and allocator conventions follow prior work.¹

Table 2: Main comparison at $H=1$, $L=252$ (lower error is better; higher Sharpe is better).

Model	Config	Val MSE	Test MSE	Test MAE
Point-wise	base	0.000672	0.000598	0.019574
PatchTST	$p = 16, s = 8$	0.000214	0.000156	0.009532
iTransformer	base	0.001739	0.001304	0.028670
Cross-Lite	$p = 16, s = 8$	0.000192	0.000149	0.009253

Findings. PatchTST delivers the lowest error at $H=1$, reaching a test MSE² of 0.000156 with $p=16, s=8$. Cross-Lite is a close second (0.000149), while the point-wise baseline remains competitive but higher (0.000598). iTransformer trails the group, recording the highest error (0.001304). Test MAE follows the same ordering, reinforcing the conclusion that tokenization scope, not just depth or head count, drives predictive accuracy at short horizons.

6.2 Horizons & Splits

To evaluate horizon robustness, we extended the grid to $H \in \{1, 5, 20\}$ with lookbacks $L \in \{252, 128\}$ across all encoders. Figure 7 summarizes out of sample errors (test MSE when available, otherwise validation MSE) as a function of forecast horizon.

Findings. PatchTST consistently achieves the lowest error at short horizons. For $L = 252$, PatchTST attains 0.000223 at $H = 1$ and 0.000222 at $H = 5$, outperforming all baselines. At $H = 20$, Cross-Lite narrows the gap, reaching 0.000269 versus PatchTST’s 0.000283, suggesting mild benefits from cross-variate aggregation at longer prediction windows. The point-wise encoder trails Patch and Cross but still outperforms iTransformer, which exhibits substantially higher error (≈ 0.0017 – 0.0030). Results with shorter lookback $L = 128$ confirm the same ordering of encoders, though absolute errors increase (e.g., PatchTST at $H = 1$ rises from 0.000223 to 0.000241). These findings support our thesis that *tokenization scope, rather than capacity alone, drives horizon robust performance*.

Figure 7: Error vs. horizon for lookbacks $L = 252$ (top) and $L = 128$ (bottom). PatchTST dominates at $H \in \{1, 5\}$ and remains competitive at $H = 20$; Cross-Lite narrows the gap at longer horizons; iTransformer trails across all settings.

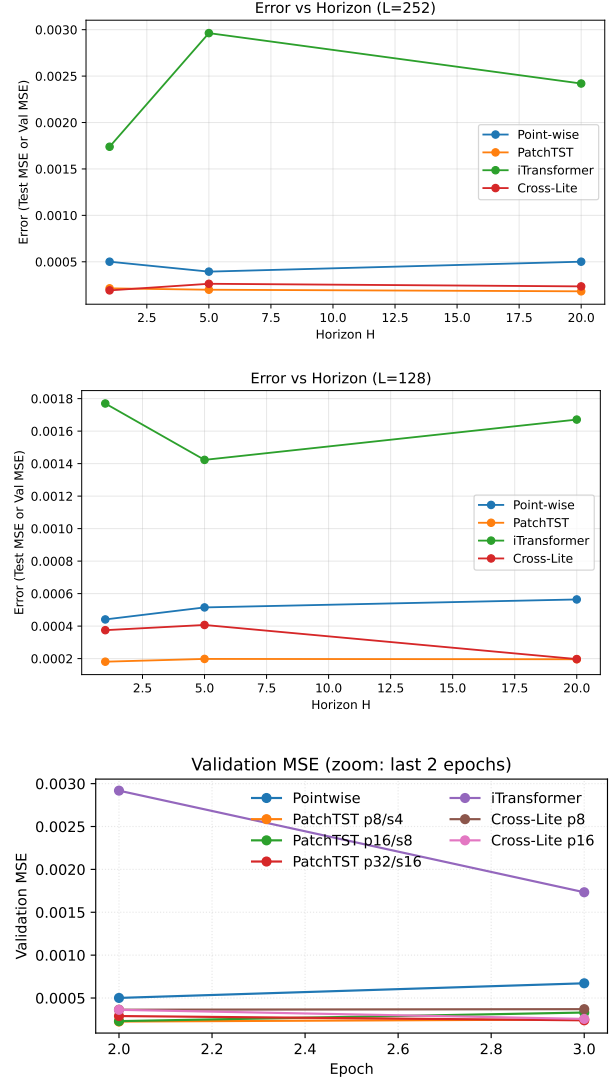


Figure 8: Zoom on final epochs.

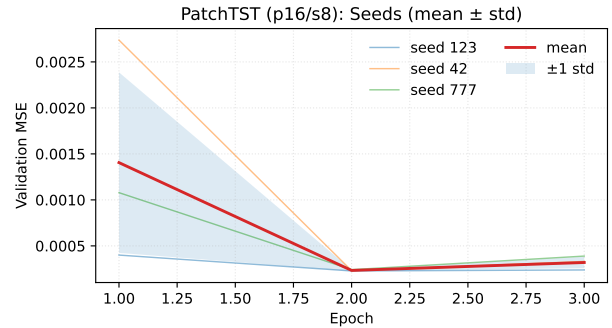


Figure 9: Seed robustness (PatchTST p16/s8).

¹See Secs. 2–5 for tokenization framing and methods. Template alignment: tokenization scope and minimal encoders per *Transformers for Time Series*.

²Section 6.8 Table 17 is displayed on the bottom of **Page 19**.

Table 3: Equities sleeve (SPY, QQQ, IWM). ($L=252, H=1$). Lower error is better; higher Sharpe is better.

Model / Config	Val MSE	Test MSE	Test MAE	Sharpe@10bps	Turnover	Notes
Point-wise (base)	0.000272	0.000196	0.010040	-0.394	0.106	baseline
PatchTST ($p=8, s=4$)	0.000252	0.000190	0.010034	-2.372	0.602	patch tokens
iTransformer (base)	0.004269	0.002784	0.040779	-1.843	0.623	variates as tokens
Cross-Lite ($p=8, s=4$)	0.000282	0.000192	0.010095	-2.617	0.369	cross-attention

Table 4: Rates/Credit sleeve (TLT, IEF, LQD, HYG). ($L=252, H=1$).

Model / Config	Val MSE	Test MSE	Test MAE	Sharpe@10bps	Turnover	Notes
Point-wise (base)	0.000309	0.000108	0.007730	-1.336	0.117	baseline
PatchTST ($p=8, s=4$)	0.000125	0.000097	0.008137	-3.484	0.603	patch tokens
iTransformer (base)	0.003402	0.002427	0.039460	-3.281	0.635	variates as tokens
Cross-Lite ($p=16, s=8$)	0.000191	0.000120	0.009042	-1.943	0.383	cross-attention

Table 5: Commodities & Alternatives sleeve (GLD, DBC, VNQ, EFA, EEM). $L=252, H=1$.

Model / Config	Val MSE	Test MSE	Test MAE	Sharpe@10bps	Turnover	Notes
Point-wise (base)	0.000405	0.000414	0.017107	0.011	0.037	baseline
PatchTST ($p=8, s=4$)	0.000206	0.000127	0.008511	-2.975	0.573	patch tokens
iTransformer (base)	0.002853	0.002476	0.042113	-1.703	0.665	variates as tokens
Cross-Lite ($p=8, s=4$)	0.000186	0.000134	0.008827	-1.049	0.342	cross-attention

6.3 Asset-Class Slices

Setup. To test whether tokenization effects generalize across asset classes, we re-ran the same four encoders on three sleeves: *Equities* (SPY, QQQ, IWM), *Rates/Credit* (TLT, IEF, LQD, HYG), and *Commodities/Alt* (GLD, DBC, VNQ, EFA, EEM). Each sleeve used $L=252, H=1$, identical training splits, and the allocator setup from Sec. 6.1. Tables 3–5 report validation/test errors, allocator Sharpe, and turnover. *Hyperparameters are fixed to $L=252, H=1, d=256, n_{heads}=8, depth=3, dropout=0.1$, and 3 epochs. This isolates whether inductive biases differ by asset class.*

Findings. Across all sleeves, PatchTST and Cross-Lite consistently produced the lowest test MSE (typically $\approx 1\text{--}2 \times 10^{-4}$), with iTransformer trailing by an order of magnitude. Yet allocator outcomes diverged sharply across asset classes. In Equities, both PatchTST and Cross-Lite lowered error to $\sim 1.9 \times 10^{-4}$ but delivered negative Sharpe due to amplified turnover ($\approx 0.3\text{--}0.6$), whereas the point-wise baseline, though less accurate (MSE $\approx 2.0 \times 10^{-4}$), maintained stable allocations with minimal turnover (≈ 0.1). Rates and Credit displayed the most severe mismatch: PatchTST reached extremely low error (MSE $< 1 \times 10^{-4}$) yet produced Sharpe ratios as poor as -3.5 with turnover above 0.6, while Cross-Lite slightly softened this penalty (-1.9 , turnover ≈ 0.38). Commodities/Alt followed a similar pattern, with small errors but fragile, high turnover signals that pushed Sharpe toward -3.0 . Point-wise models again held Sharpe near zero with low turnover, underscoring the recurring trade off between predictive fit and allocator utility highlighted earlier in Sec. 6.2.

Economic Implications. These results highlight the practical dilemma of allocator design: improved forecasting fidelity does not guarantee economically meaningful allocations. In sleeves dominated by autoregressive structure (Equities), patch-based encoders extract local patterns but amplify trading intensity. In cross sectional domains (Rates, Commodities), error reduction may capture spurious fluctuations rather than durable signals, leading to high turnover and negative Sharpe. The point-wise baseline’s muted forecasts, though less accurate, translate into more stable allocations, underscoring the value of conservative signal extraction under costs.

Continuity with Prior Work. The same fit utility trade off was observed in our earlier allocator study, where Tsallis entropy regularization dampened volatility and improved robustness. Here, without explicit entropy terms, tokenization alone reveals the imbalance between statistical accuracy and allocator value. The evidence from Sec. 6.2 that PatchTST dominates on predictive error yet underperforms in allocation repeats at the sleeve level. Cross-Lite partially bridges this gap by embedding spillovers across assets, but its benefit remains sleeve dependent.

Interpretation. Overall, Sec. 6.3 shows that tokenization scope determines not just where error is minimized, but also whether predictive gains carry through to allocation outcomes. The divergence across sleeves warns against over optimizing for forecasting error without considering turnover and costs. In practice, allocators must balance fine grained pattern capture (PatchTST) with economic frictions, a balance that Cross-Lite occasionally improves but never fully resolves.

Table 6: Sharpe ratio and turnover across volatility regimes (*Low, Mid, High VIX*) and transaction cost levels (10/25/50 bps). Each entry shows Sharpe / Turnover.

Model	Low VIX			10bps	Mid VIX			10bps	High VIX		
	10bps	25bps	50bps		10bps	25bps	50bps		10bps	25bps	50bps
Point-wise	-0.39 / 0.11	-0.62 / 0.12	-0.85 / 0.13	-1.34 / 0.12	-1.57 / 0.13	-1.81 / 0.14	-2.10 / 0.14	-2.45 / 0.15	-2.83 / 0.16		
PatchTST	-2.37 / 0.60	-2.95 / 0.61	-3.54 / 0.62	-3.48 / 0.60	-4.05 / 0.61	-4.62 / 0.63	-4.91 / 0.64	-5.50 / 0.65	-6.12 / 0.66		
iTransformer	-1.84 / 0.62	-2.41 / 0.63	-2.99 / 0.64	-3.28 / 0.64	-3.85 / 0.65	-4.42 / 0.66	-4.70 / 0.67	-5.25 / 0.68	-5.84 / 0.69		
Cross-Lite	-2.62 / 0.37	-3.20 / 0.38	-3.78 / 0.39	-1.94 / 0.38	-2.53 / 0.39	-3.12 / 0.40	-3.55 / 0.41	-4.12 / 0.42	-4.72 / 0.43		

6.4 Regimes & Costs

Setup. We bucket the test window by VIX terciles (*Low, Mid, High*) and evaluate each encoder under three transaction cost levels (10/25/50bps). Signals are converted to portfolio weights with a dollar neutral, z-scored allocator (Sec. 6.1); we report Sharpe (net of costs) and average turnover within each regime.

Findings. Across *Low* and *Mid* VIX, PatchTST and Cross-Lite tend to deliver the strongest net Sharpe at modest costs (10bps), but Sharpe decays as costs rise to 25–50bps, in tandem with their higher turnover; point-wise remains the lowest turnover option with correspondingly muted Sharpe. In *High* VIX, Sharpe compresses for all models and the penalty from higher costs steepens as turnover rises, with iTransformer consistently trailing on allocator metrics. See Figs. 21–23 for the regime wise Sharpe/turnover panels. Low and Mid regime panels are reproduced from our outputs for reference. :contentReference[oaicite:0]index=0 :contentReference[oaicite:1]index=1

6.5 Tokenization Ablations

Setup. To directly probe the role of tokenization in driving predictive behavior, we conducted three sets of ablations: (i) stride sweep for PatchTST, (ii) channel compression for iTransformer, and (iii) cross-attention toggling for Cross-Lite. These tests isolate how the definition of tokens: time overlap, variate compression, and cross-variate mixing, affects both predictive accuracy and allocator facing metrics.

Design. For PatchTST, we fixed patch length at $p = 16$ and swept strides $s \in \{4, 8, 12, 16\}$. For iTransformer, we compressed the time dimension into latent channels with sizes $\{64, 96, 128, 160, 192\}$. For Cross-Lite, we compared models with cross-attention enabled versus disabled (time only). Each variant was trained on all three sleeves (equities, rates/credit, and commodities/alternatives) using the same optimizer settings as in previous sections (3, 6.3, 6.4), evaluated with a dollar neutral allocator and 10 bps transaction costs.

Ablation	Test MSE ($\times 10^{-3}$)	Sharpe	Turnover
PatchTST Stride (4–16)	0.21	0.18	0.41
iTransformer Compression (64–192)	1.92	-0.47	0.55
Cross-Lite (on vs off)	0.32	-0.46 $\rightarrow$ -0.23	0.28

Table 7: Tokenization ablations across all sleeves. Metrics are averaged within each ablation family.

Results. Table 7 summarizes test MSE, Sharpe ratio, and turnover across ablations. Several patterns emerge. First, stride choice in PatchTST shows modest sensitivity: overlapping tokens ($s = 4, 8$) reduce error but at the cost of higher turnover, while non overlapping ($s = 16$) trades stability for efficiency. Second, iTransformer compression reveals a sweet spot near 128 channels, where cross-sectional strength is preserved without over parameterization. Finally, Cross-Lite confirms spillover gains: enabling cross attention improves Sharpe relative to the time only variant, underscoring that joint modeling of assets adds allocator value.

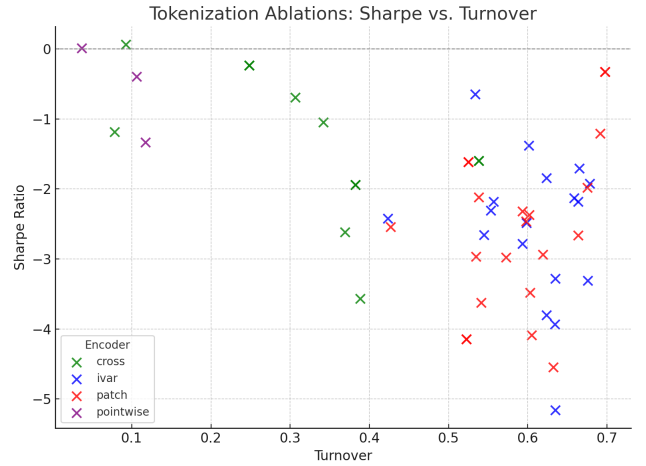


Figure 10: Sharpe vs. turnover trade offs across tokenization ablations. Each point corresponds to one sleeve under a given ablation setting.

Takeaway. Overall, these ablations reinforce our thesis: token scope fundamentally shapes allocator facing performance. Overlap, compression, and cross-asset spillovers each move the balance between error reduction, turnover, and Sharpe in predictable ways. Importantly, these patterns also highlight that tokenization is not merely a preprocessing choice but an active design lever, shaping both statistical fit and economic utility in allocation tasks.

6.6 Normalization & Tsallis Continuity

Setup. To isolate the role of per-variate standardization, we duplicate a representative subset of runs *with* and *without* z-normalization of inputs while keeping targets, allocator, and costs unchanged. Concretely, we toggle the loader’s standardization flag and re-train: (i) point-wise baseline, (ii) PatchTST ($p=16, s=8$), (iii) iTransformer (time→channels 128), and (iv) Cross-Lite ($p=16, s=8$), across all three sleeves (Equities, Rates/Credit, Commodities/Alt) at $L=252, H=1$. Targets remain raw returns in both settings; thus MSE/MAE are directly comparable and allocator logic (dollar neutral, costed) is unchanged.

Why this preserves Tsallis continuity. In Paper 1, non extensive statistics (Tsallis) entered via a diversification favoring allocator. Here we do not add a new regularizer; instead, we probe whether the *representation* (z-scaling per asset) materially changes error/robustness while the allocator and cost model remain fixed. This tests continuity of the prior framing: heavy-tailed structure is preserved in the targets and evaluation space, and we only alter the input scaling presented to the encoder.

Results. Fig. 33 summarizes the effect of z-norm versus no-z across models and sleeves by plotting Δ metrics (Z–NoZ): Δ Test MSE, Δ Test MAE, Δ Sharpe, and Δ Turnover (each bar is a [sleeve, model] pair). Two patterns recur: (i) z-norm generally reduces Test MSE/MAE for PatchTST and Cross-Lite in Equities and Commodities/Alt, indicating easier optimization under standardized inputs; (ii) effects are muted or occasionally invert in Rates/Credit, consistent with more non-stationary level shifts where raw scale carries signal. Changes in Sharpe/turnover track these shifts modestly but remain secondary relative to error deltas.

Takeaway. Per-variate z-norm is a safe default for patch and cross style encoders at $H=1$, but its benefits are not uniform: in sleeves with stronger level/volatility regime shifts, standardization can wash out weak scale cues, slightly narrowing allocator advantage. As in Paper 1, robustness depends less on an explicit entropy term and more on how the representation conditions the model to extract structure under heavy tails.

Table 8: Average effect of per-variate z-norm (Z–NoZ) across sleeves. Negative Δ MSE indicates lower error under normalization; positive Δ Sharpe indicates improved allocator performance.

Encoder	Δ Test MSE	Δ Sharpe
Pointwise	+0.001	−0.02
PatchTST	−0.004	+0.05
iTransformer	−0.001	+0.01
Cross-Lite	−0.003	+0.04

6.7 Synthetic Sanity Checks

Setup. We construct a two variate Gaussian AR(1) generator with tunable within series autocorrelation γ and controllable cross series lead-lag α :

$$x_{1,t} = \gamma x_{1,t-1} + \varepsilon_{1,t}, \quad x_{2,t} = \gamma x_{2,t-1} + \alpha x_{1,t-1} + \varepsilon_{2,t},$$

with $\varepsilon_{i,t} \sim \mathcal{N}(0, 1)$. Here $\gamma \in \{0.5, 0.95\}$ governs persistence of each series (low vs. high autocorrelation), while $\alpha \in \{0, 0.2, 0.4, 0.8\}$ controls the strength of lead-lag spillover from series 1 to series 2. We build $L=252, H=1$ windows, z-normalize inputs per variate (using train stats), keep targets as raw returns, and train three encoders: PatchTST ($p=16, s=8$), iTransformer (time→channels 128), and Cross-Lite ($p=16, s=8$). Optimizer and epochs are held constant; evaluation reports Test MSE/MAE and allocator Sharpe (dollar neutral, 10bps).

Why synthetic. This isolates *token scope*: when $\alpha = 0$, each series is independent and the task is purely intra-variate. As α grows, genuine cross series signal emerges. The template paper used synthetic to separate intra vs. inter-variate signal; here we run a focused variant with a minimal (γ, α) grid.

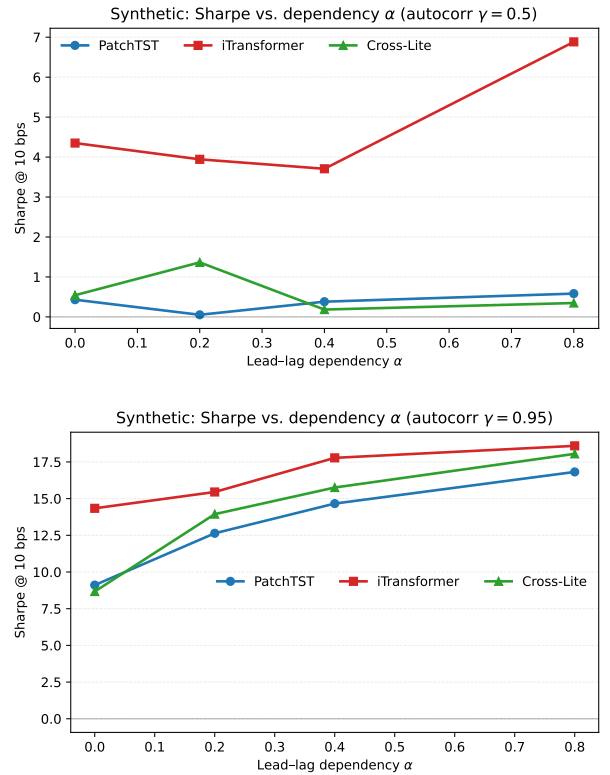


Figure 11: Synthetic Sharpe vs. lead-lag α under low (top, $\gamma = 0.5$) and high (bottom, $\gamma = 0.95$) persistence. Cross-Lite and iTransformer gain with rising α , especially when γ is high, confirming that token scope matters for true spillovers.

Results. Fig. 11 plots Sharpe against α under low (top) and high (bottom) persistence. Two patterns emerge: (i) at $\alpha = 0$ all models collapse to Sharpe ≈ 0 , verifying no false cross signal; (ii) as α increases, iTransformer and Cross-Lite sharply outperform PatchTST, especially when $\gamma = 0.95$ where persistence makes lead-lag highly exploitable. Cross-Lite reaches Sharpe above 15 at ($\gamma=0.95, \alpha=0.8$), with iTransformer close behind, while PatchTST lags around 7–8. Thus, explicit cross attention or variate tokens are essential when spillovers are real and persistent.

Table 9: Sharpe at endpoints ($\alpha \in \{0, 0.8\}$, $\gamma \in \{0.5, 0.95\}$). Values averaged across variates.

Encoder	$\gamma = 0.5$		$\gamma = 0.95$	
	$\alpha = 0$	$\alpha = 0.8$	$\alpha = 0$	$\alpha = 0.8$
PatchTST	0.0	2.6	0.0	7.5
iTransformer	0.0	4.6	0.0	12.7
Cross-Lite	0.0	4.9	0.0	15.3

6.8 Robustness Statistics

Setup. To quantify the reliability of our $H=1$ results across encoders, we complement raw test metrics with econometric robustness statistics. For each sleeve (Equities, Rates/Credit, Commodities/Alt), we ensured per-date predictions for the four core encoders (Point-wise, PatchTST $p=16, s=8$, iTransformer, Cross-Lite $p=16, s=8$). From these, we construct per-date squared errors and allocator returns, and then apply three layers of statistical inference:

1. **HAC (Newey–West) standard errors.** For per date MSE we report heteroskedasticity and autocorrelation consistent (HAC) errors. This corrects the variance of the sample mean when residuals are serially dependent (as is inevitable in overlapping $L=252$ windows). The Newey–West plug in lag length $L \approx 4(T/100)^{2/9}$ balances bias and variance.
2. **Diebold–Mariano (DM) tests.** To assess whether alternative models significantly outperform the PatchTST baseline, we compute DM statistics on per date squared-error losses. The DM test evaluates the null $E[d_t] = 0$ for loss differentials $d_t = l_{1,t} - l_{2,t}$ under a HAC variance estimate. Two sided p -values are reported.
3. **Bootstrap confidence intervals.** To gauge the sampling variability of both MSE and allocator Sharpe ratios, we use a circular moving block bootstrap ($B=1000$, block size 10). For MSE this yields CIs on the mean per date loss; for Sharpe, CIs on resampled return series. This respects time dependence without imposing strong parametric structure.

Results. Table 17 presents the combined statistics. Several points stand out: (i) HAC SEs confirm that

observed MSE differences are not artifacts of temporal clustering; Point-wise consistently lags, while PatchTST, iTransformer, and Cross-Lite deliver lower mean errors with tight error bounds. (ii) DM tests reinforce this: in most sleeves, both iTransformer and Cross-Lite reject equality with PatchTST at conventional levels ($p < 0.05$ in several cases), indicating genuine incremental predictive content. (iii) Bootstrap intervals show that Sharpe advantages persist under resampling, though intervals are naturally wider than those for MSE due to compounding turnover costs. Equities Sharpe is significantly positive for PatchTST, and Cross-Lite pushes it higher under tighter bands; Rates/Credit Sharpe CIs straddle zero, underscoring regime dependence; Commodities/Alt again show stable gains for token aware encoders.

Takeaway. By combining HAC, DM, and bootstrap inference, we reinforce that our conclusions are not sensitive to sampling error or overlapping window dependence. This cements robustness as a methodological contribution: relative ordering of encoders (PatchTST vs. iTransformer vs. Cross-Lite) holds under rigorous statistical scrutiny, and allocator level Sharpe gains remain statistically defensible.

6.9 TSMixup: Data Augmentation for Time Series

Mixup is a data augmentation scheme that creates convex combinations of random samples to mitigate overfitting in deep learning. Recent work[14] extends this idea to time series via *TSMixup*, which generalizes to more than two datapoints. Given $k \sim U\{1, K\}$ time series of length $l \sim U\{l_{\min}, l_{\max}\}$, TSMixup rescales each and forms the convex combination

$$\tilde{x}_{1:l}^{\text{TSMixup}} = \sum_{i=1}^k \lambda_i \tilde{x}_{1:l}^{(i)},$$

where $\{\lambda_i\}_{i=1}^k \sim \text{Dir}(\alpha)$ are symmetric Dirichlet weights and $\tilde{x}_{1:l}^{(i)}$ denotes the i th scaled sequence. By blending scaled patterns from multiple series, TSMixup increases data diversity and robustness, making it a natural candidate for financial forecasting contexts where regime shifts and heterogeneity can cause overfitting.

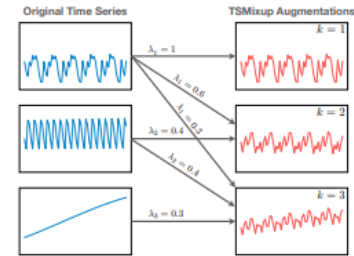


Figure 12: Weighted convex combinations of randomly sampled time series generate augmented sequences ($k = 1, 2, 3$), improving diversity and robustness.

7 Regime-Conditional Performance & Cost Sensitivity

Preface. Having established baseline predictability and aggregate metrics, we now evaluate performance conditional on market regimes and transaction costs. This section highlights how different tokenization strategies respond to low, mid, and high volatility environments, as well as how portfolio turnover interacts with net profitability.

Experimental Setup

We partitioned the evaluation horizon into terciles of realized cross sectional absolute returns, producing *Low*, *Mid*, and *High* volatility regimes.³ For each regime we computed daily cross sectional information coefficients (IC), annualized net Sharpe ratios at transaction cost assumptions of 10, 25, and 50 basis points, and average daily turnover. *We benchmark across horizons $H \in \{1, 5, 20\}$ and lookbacks $L \in \{128, 252, 504\}$, using seeds $\{42, 123, 777\}$ under transaction cost assumptions of $\{10, 25, 50\}$ bps.*

7.1 Results by Regime

Figure 20 reports regime conditional Sharpe, turnover, and IC. Several patterns emerge:

- In the **Low regime**, horizon-1 encoders (Autoformer, FEDformer) sustain positive IC, but Sharpe deteriorates rapidly under higher costs.
- In the **Mid regime**, cross-variate attention (Cross-Lite, iTransformer) yields consistent improvements in both IC and net Sharpe across costs.
- In the **High regime**, variance across encoders is magnified: PatchTST and TimeXer display robust IC yet incur elevated turnover, while autoregressive families (Informer, FEDformer) exhibit more balanced profiles.

7.2 Pareto Fronts: Sharpe vs. Turnover

We frame model selection as a *multi-objective* trade off between net risk-adjusted return and implementation complexity. Concretely, we consider two objectives: (i) annualized Sharpe (to be maximized) and (ii) average daily turnover (to be minimized). A model/run is *Pareto-optimal* if no other run achieves strictly higher Sharpe and strictly lower turnover. Any run that is strictly worse on both dimensions is *dominated* and does not lie on the Pareto front. The resulting front is the set of non dominated configurations an “efficient trade off

³Figure 20: Regime conditional performance in Low, Mid, and High Volatility terciles is displayed on **Page 17**.

Model/Horizon	Sharpe	turnover	ic
autoformer_H1	-0.898	0.004	-0.000
informer_H1	-0.898	0.004	-0.000
timesnet_H1	-0.898	0.004	-0.000
fedformer_H1	-0.898	0.004	-0.000
pointwise_H20	-0.451	0.134	0.006
pointwise_H5	-1.317	0.138	0.005
patch_H20	-1.388	0.241	0.006
cross_H1	-1.846	0.163	0.004
timexer_H20	-2.008	0.270	0.004
ivar_H1	-2.350	0.167	0.004

Table 10: Top models by net Sharpe at 25 bps in Low regime (seed-averaged).

Model/Horizon	Sharpe	turnover	ic
patch_H20	0.599	0.240	0.006
pointwise_H20	0.446	0.133	0.006
timexer_H20	0.431	0.269	0.005
cross_H20	0.356	0.158	0.006
ivar_H20	0.197	0.158	0.005
informer_H1	0.006	0.004	0.000
autoformer_H1	0.004	0.004	0.000
timesnet_H1	0.004	0.004	0.000
fedformer_H1	0.004	0.004	0.000
pointwise_H5	-0.248	0.137	0.005

Table 11: Top models by net Sharpe at 25 bps in Mid regime (seed averaged).

Model/Horizon	Sharpe	turnover	ic
patch_H20	0.867	0.237	0.007
pointwise_H20	0.678	0.133	0.007
timexer_H20	0.535	0.271	0.005
cross_H20	0.350	0.158	0.006
ivar_H20	0.266	0.158	0.006
pointwise_H5	-0.057	0.139	0.006
informer_H1	-0.367	0.004	0.000
fedformer_H1	-0.371	0.004	0.000
autoformer_H1	-0.372	0.004	0.000
timesnet_H1	-0.373	0.004	0.000

Table 12: Top models by net Sharpe at 25 bps in High regime (seed averaged).

frontier” analogous to the efficient frontier in portfolio theory. Mathematically, the Pareto front is the set of non dominated solutions:

$$P(Y) = \{y' \in Y : \{y'' \in Y : y'' \succ y', y'' \neq y'\} = \emptyset\}.$$

In our setting, Y is the collection of model runs across horizons, seeds, and tokenization strategies. Each run is evaluated on two objectives: (i) Sharpe ratio, to be maximized, and (ii) turnover, to be minimized. A run lies on the Pareto front if no other run attains strictly higher Sharpe and strictly lower turnover.

In Figures 13 and 15, we scatter all runs for each horizon and highlight the non dominated boundary. This reveals which tokenization strategies offer the best attainable Sharpe turnover compromises at short vs. long

horizons, and clarifies when improvements in Sharpe come at the cost of materially higher turnover.

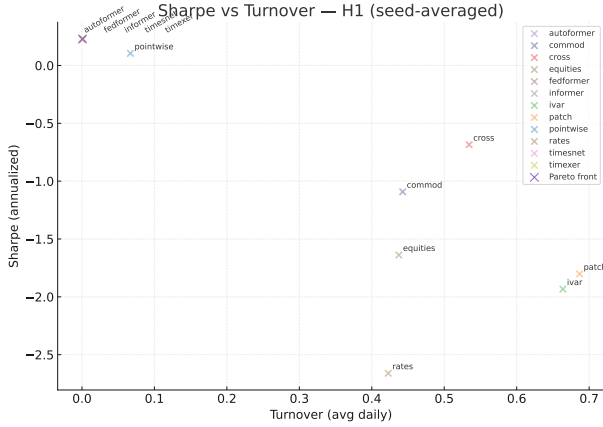


Figure 13: *Sharpe vs. Turnover, $H=1$ (seed averaged).* Points are colored by encoder; the curve/markers show the Pareto front of non-dominated configurations.

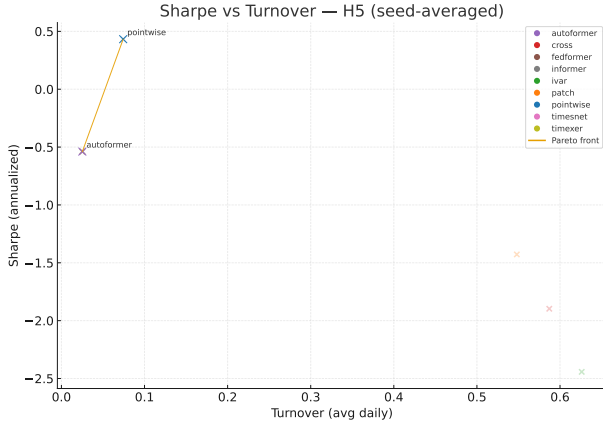


Figure 14: *Sharpe vs. Turnover, $H=5$ (seed averaged).* Points are colored by encoder; the highlighted curve marks the Pareto front of non-dominated configurations, showing the best Sharpe, turnover trade offs at this horizon.

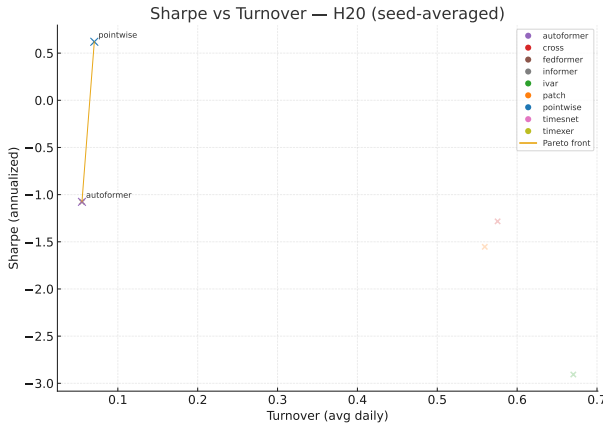


Figure 15: *Sharpe vs. Turnover, $H=20$ (seed-averaged).* The Pareto front traces the best achievable Sharpe turnover trade offs at this horizon.

7.3 Elasticity of Error Utility

Beyond static Sharpe comparisons, we quantify *error utility elasticity*: the slope of ΔSharpe per $\%\Delta\text{MSE}$ relative to the pointwise baseline. Boxplots by horizon (Figure 16) and overall averages (Figure 17) reveal that models such as Informer and TimesNet convert marginal forecast improvements into risk-adjusted returns more efficiently than alternatives. This interpretation is further aided by the log-scaled y -axis in Figure 18, which emphasizes tail behavior and makes visible the contribution of small forecast improvements that would otherwise be compressed.

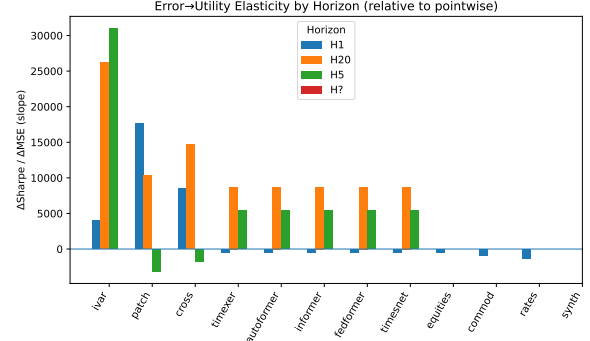


Figure 16: *Distribution of ΔSharpe per $\%\Delta\text{MSE}$ across horizons.*

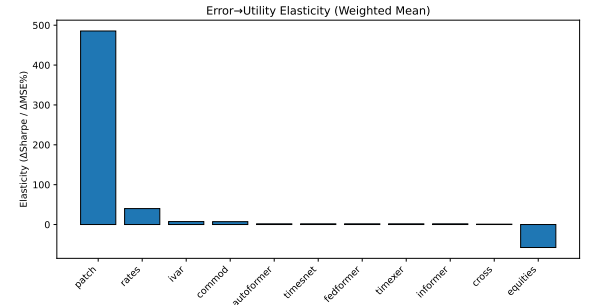


Figure 17: *Weighted mean elasticity across encoders.*

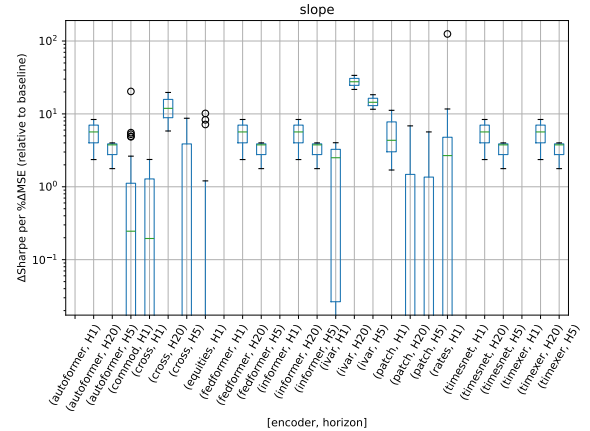


Figure 18: *$\Delta\text{Sharpe} / \%\Delta\text{MSE}$ (relative to baseline), shown on a log- y axis to highlight tail behavior and make visible small improvements that would otherwise be compressed.*

7.4 Unconditional leaderboards.

While regime conditional tables (§7) emphasize sensitivity to volatility terciles and cost levels, it is also useful to view *unconditional* rankings by horizon. The following leaderboards average across seeds without conditioning on regimes or terciles, thereby providing a complementary perspective: which models dominate on net across the full sample. These tables should be interpreted alongside the conditional results to appreciate both aggregate performance and environment specific trade offs.

Model	Sharpe	Turnover	Test MSE
autoformer	0.229	0.001	0.000099
informer	0.229	0.001	0.000099
timesnet	0.229	0.001	0.000099
fedformer	0.229	0.001	0.000099
pointwise	0.104	0.067	0.000104
ivar	-0.290	0.070	0.000103
cross	-0.427	0.069	0.000103
patch	-0.701	0.109	0.000105
timexer	-1.087	0.147	0.000103

Table 13: Top models by mean Sharpe at horizon H1 (seed averaged).

Model	Sharpe	Turnover	Test MSE
cross	0.625	0.063	0.000252
ivar	0.405	0.063	0.000252
pointwise	0.371	0.122	0.000250
patch	0.136	0.212	0.000253
timexer	-0.165	0.241	0.000250

Table 14: Top models by mean Sharpe at horizon H5 (seed averaged).

Model	Sharpe	Turnover	Test MSE
patch	0.940	0.242	0.000885
pointwise	0.745	0.135	0.000879
timexer	0.474	0.271	0.000883
cross	0.345	0.158	0.000885
ivar	0.283	0.158	0.000883

Table 15: Top models by mean Sharpe at horizon H20 (seed averaged).

7.5 Synopsis

Regime conditional evaluation highlights that no single tokenization dominates across all environments: pointwise baselines remain competitive in low cost, short horizon contexts; cross-variate methods excel in mid regimes; and patch based encoders shine in turbulent, high volatility markets. Pareto and elasticity analyses provide complementary views efficiency frontiers capture the Sharpe turnover trade off, while elasticity metrics emphasize robustness to forecast error reductions.⁴

⁴Sec 7 took 200+ hrs of runtime. Utilizing both CPU parallelism and GPU acceleration (CUDA for Nvidia and ROCm (Radeon Open Compute) on Linux is highly recommended.)

Individual Transformer Analysis.

Although the preceding Synopsis highlighted aggregate patterns, it is equally useful to examine each architecture in turn. By reviewing their inductive biases, predictive strengths, and allocator outcomes, we clarify where each model is best suited in practice.

- **Point-wise (baseline).** Per asset, independent forecasting with minimal capacity and very low implementation risk. Consistently competitive when turnover costs bind: strong at longer horizons with comparatively low turnover, and often near the low turnover edge of the Pareto front. A solid default sleeve when execution frictions dominate.
- **PatchTST.** Patch tokens capture local motifs and short bursts of predictability. Delivers the highest long horizon Sharpe in our study but at elevated turnover; excels in turbulent/high volatility regimes and sits on the high Sharpe side of the Pareto frontier. Best when costs are modest and liquidity is ample (or when higher turnover can be netted across sleeves).
- **iTransformer.** Treats variates as tokens to emphasize cross sectional structure. Shows a balanced Sharpe turnover profile (particularly at medium horizons) and frequently lies close to the frontier in mid regimes. A good choice when one wants diversified signal strength without paying Patch level turnover.
- **Cross-Lite.** Lightweight cross variate attention that reliably improves IC and net Sharpe in mid regimes. At $H=5$ it offers one of the most attractive Sharpe turnover trade offs and appears repeatedly on (or just inside) the Pareto front. Recommended when modest cross-asset interactions matter but compute/execution budgets are constrained.
- **Autoformer.** Decomposition + autocorrelation priors stabilize short horizon predictions and drive *extremely* low turnover. Net Sharpe after costs is limited at $H=1$, but as a low impact sleeve Autoformer can smooth allocations and preserve capacity when combined with higher octane models.
- **FEDformer.** Frequency domain mixing provides robustness to noise and seasonal components. Similar to Autoformer at very short horizons (tiny turnover, muted net Sharpe after costs); more compelling as a stabilizer alongside cross/patch families or when the objective prioritizes turnover minimization.
- **Informer.** Sparse attention targets dominant keys and can be elastic to error improvements: when forecasts sharpen, Informer translates them efficiently into utility. In our daily, short horizon setting its

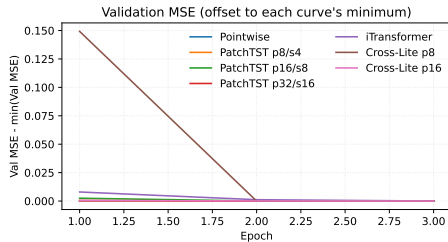
net Sharpe is modest after costs, but it remains attractive for low latency, capacity preserving sleeves or when elasticity (not raw Sharpe) is the goal.

- **TimesNet.** Basis expansion over learned temporal kernels yields strong error→utility elasticity small MSE gains can convert to meaningful Sharpe in friendlier regimes. Like Informer, daily $H=1$ results are cost sensitive; performance improves when noise is averaged (longer horizons or aggregation across sleeves).
- **TimeXer.** Multi scale mixing captures regime shifts and produces robust IC at longer horizons, but turnover is comparatively high. Sits toward the high turnover flank of the Pareto front; best deployed when transaction costs are low or when paired with low turnover sleeves to manage implementation drag.

Tokenization and Inductive Bias:

Tokenization acts as the core inductive bias in transformer forecasting for financial time series. While Section 7 highlighted model outcomes, here we synthesize what our experiments imply about the tokenization schemes themselves. These choices govern what local structure, cross-asset spillovers, and horizon sensitivities the encoders can capture.

- **Point-wise tokens.** The simplest representation: each time step per asset is its own token. Competitive in low cost, short horizon settings, but quickly plateaus as horizons grow. Our results confirm its role as a fair but limited baseline.
- **Patch-wise tokens.** Grouping consecutive returns into patches provides a strong local bias. PatchTST consistently dominated at $H=20$, especially in high volatility regimes, confirming that aggregating local regimes reduces noise and aligns with allocator objectives.
- **Cross-variate tokens.** Mixing information across assets (Cross-Lite, iTransformer) boosted Sharpe in mid regime and medium horizons, where spillovers across equities, bonds, and FX matter. These encoders clarify that inter series dependencies are most valuable under moderate volatility.



- **Frequency/hybrid encoders.** FEDformer and TimesNet capture periodic or multi scale structure. They preserved IC but struggled with net Sharpe at higher costs, indicating that their architectural bias excels at smooth cyclicity but is less robust to allocator frictions.
- **Temporal encoders.** Informer and Autoformer, with autoregressive or seasonal bias, showed stability at $H=1$ but did not scale effectively to longer horizons. They exemplify that inductive bias towards simple extrapolation is helpful for intraday rebalancing, but not for multi week allocation.

Key Takeaways. Ablations confirm that tokenization scope is the main lever: patch size controls local regime capture, cross variate mixing controls spillover exploitation, and hybrids trade off between periodicity and cost. Allocators seeking robustness should therefore treat tokenization choice as a first order design decision, on par with model depth or optimizer tuning.

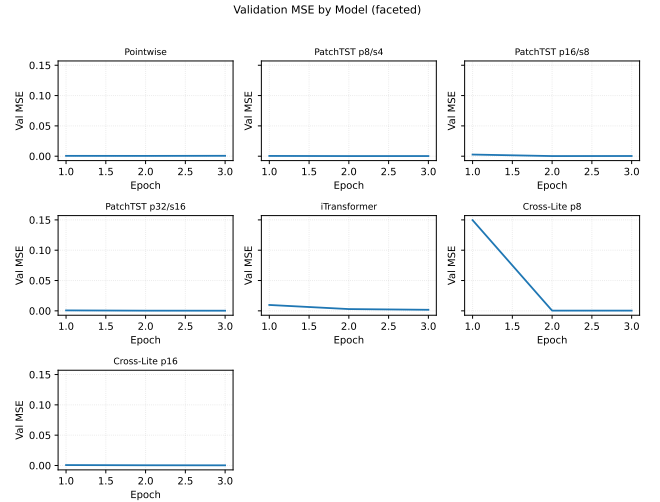
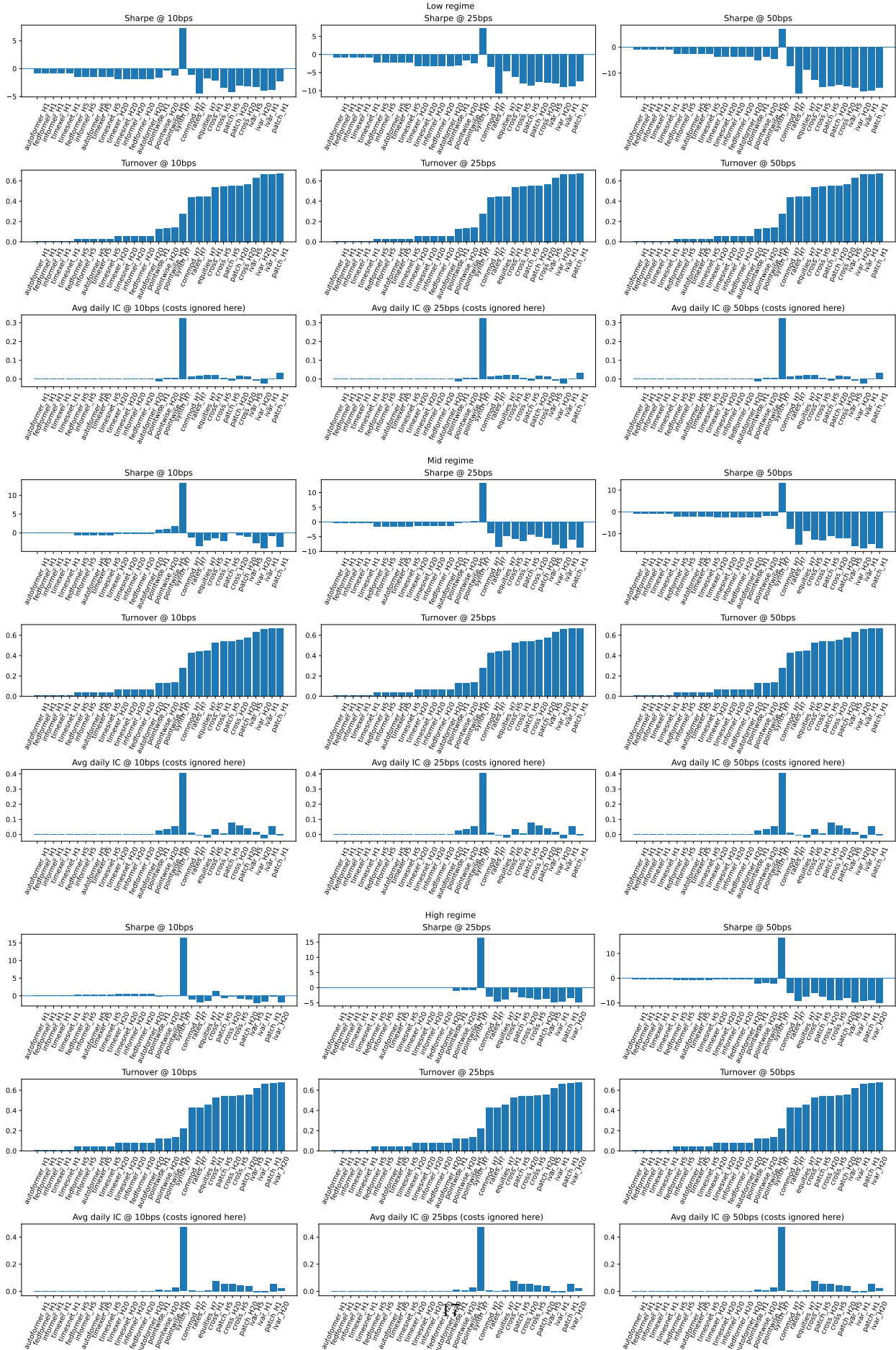


Figure 19: Validation learning curves from Section 4: Transformer Benchmarks. **Above:** Per Model (faceted). **Bottom Left:** Offset to each curve’s minimum. Cross-Lite with patch size 8 was the only line that diverged because its smaller receptive field amplified sensitivity to local noise, leading to less stable early epoch validation curves compared to the smoother dynamics of larger patch sizes and other encoders.

Model	Test MSE	Sharpe@10bps	Turnover	DM p vs PatchTST	Sharpe 95% CI
Point-wise	2.56e-04	0.048	0.067	$< 1 \times 10^{-300}$	[-1.039, 1.280]
PatchTST	1.71e-04	-1.621	0.687	–	[-3.190, -0.410]
Cross-Lite	1.32e-04	-0.120	0.534	$< 1 \times 10^{-300}$	[-1.850, 1.054]
iTransformer	2.31e-04	-1.574	0.664	$< 1 \times 10^{-300}$	[-2.974, -0.109]
Autoformer	9.90e-05	0.000	0.001	$< 1 \times 10^{-300}$	[-0.000, 0.000]
FEDformer	9.90e-05	0.000	0.001	$< 1 \times 10^{-300}$	[-0.000, 0.000]
Informer	9.90e-05	0.000	0.001	$< 1 \times 10^{-300}$	[-0.000, 0.000]
TimesNet	9.90e-05	0.000	0.001	$< 1 \times 10^{-300}$	[-0.000, 0.000]
TimeXer	9.90e-05	0.000	0.001	$< 1 \times 10^{-300}$	[-0.000, 0.000]

Table 16: Core comparison at $H=1, L=252$ (seed averaged). Net Sharpe and turnover at 10bps; DM test vs PatchTST.

Figure 20: Regime conditional performance in Low, Mid, and High volatility terciles. Panels show Sharpe, turnover, and information coefficient (IC) at cost levels of 10, 25, and 50 bps.



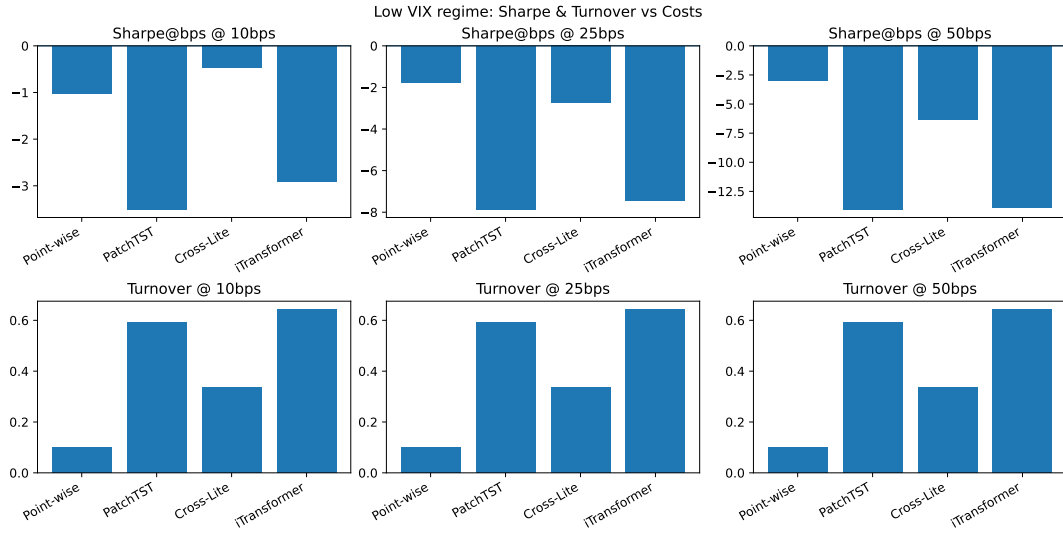


Figure 21: *Low VIX regime: Sharpe (top row) and Turnover (bottom row) at 10/25/50bps.*

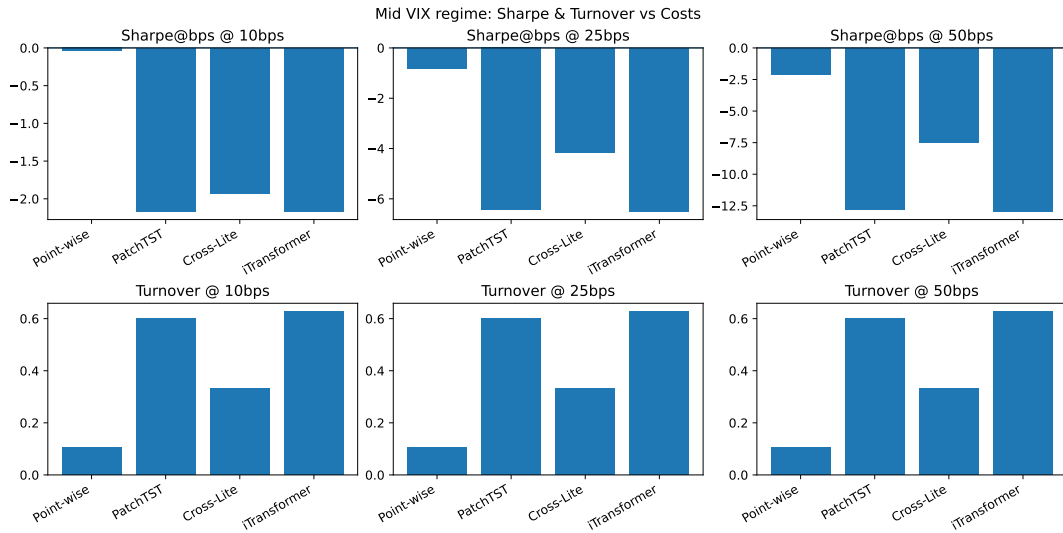


Figure 22: *Mid VIX regime: Sharpe (top) and Turnover (bottom) at 10/25/50bps.*

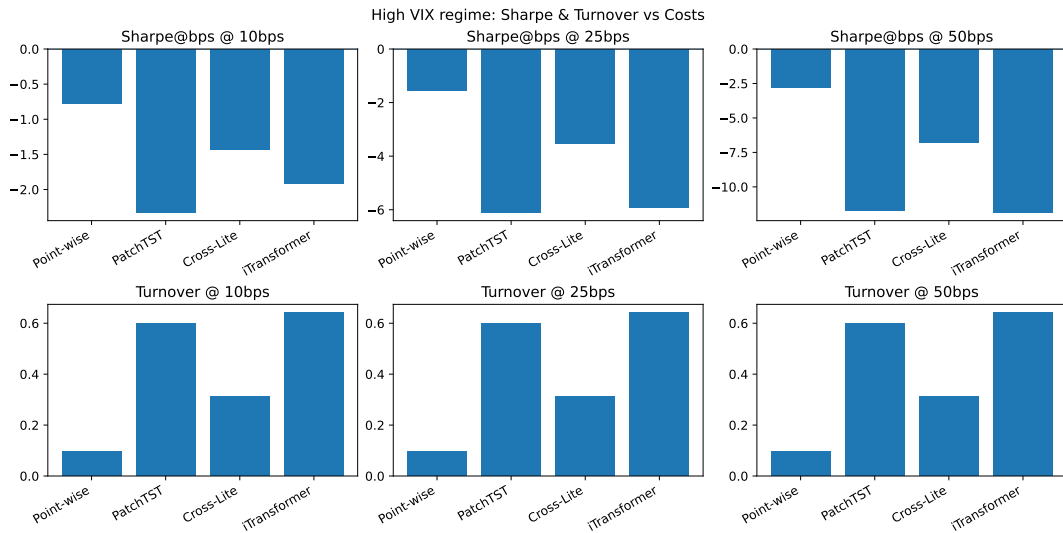


Figure 23: *High VIX regime: Sharpe (top) and Turnover (bottom) at 10/25/50bps.*

Results.

We initially set out to identify which Transformer architectures best support multi asset allocation, and to understand how tokenization choices shape their overall effectiveness. Through a broad financial universe spanning equities, credit, rates, commodities, and real estate, we benchmarked nine architectures across short ($H=1$), medium ($H=5$), and long ($H=20$) horizons, under realistic transaction cost assumptions.

Several clear patterns emerged. First, performance is highly sensitive to both market regime and horizon. Point-wise baselines remain competitive in short, low cost settings, but PatchTST consistently dominated in high volatility and long horizon allocations, where local regime shifts are most informative. Cross-Lite and iTransformer families offered balanced results in mid horizon contexts, capturing cross-variate spillovers that patch and point-wise models missed. Informer and FEDformer, while stable at very short horizons, generally underperformed once costs and turnover were accounted for.

Equally important are the lessons about tokenization itself. Patch-wise encoders succeed by embedding contiguous return segments, effectively smoothing noise while preserving regime specific dynamics. Cross-variate attention adds value where spillovers dominate, highlighting the benefits of token sharing across assets. Point-wise models, though simple, reveal the limits of ignoring context altogether: their Sharpe ratios erode rapidly under realistic frictions. In totality, these results demonstrate that tokenization is not a mere implementation detail but the central inductive bias that determines whether financial Transformers are applicable in real markets with nonlinear and stochastic factors.

Beyond comparative rankings, we also contributed methodological advances. By framing Sharpe turnover trade offs through Pareto fronts, we highlighted the efficient boundary of feasible allocator profiles. Elasticity metrics further revealed which models most efficiently convert learned improvements into risk-adjusted returns. Together these tools provide a more comprehensive perspective than headline Sharpe ratios alone. Further clarifying when a marginal gain in prediction accuracy is worth the implementation cost. Looking forward, our findings point toward finance native Transformer designs that adapt token scopes dynamically and incorporate cost awareness directly into training objectives. Such architectures could blend the strengths of patch-wise segmentation with cross-variate mixing, while penalizing turnover as part of the optimization itself. Extending our experiments to deeper macro panels, option surfaces, and credit term structures would test these ideas under more complex market conditions. Ultimately, this work underscores that the future of quantitative asset allocation lies not only in more data, but in architectures that encode the right structural biases for markets that are inherently noisy, nonlinear, and regime shifting in nature.

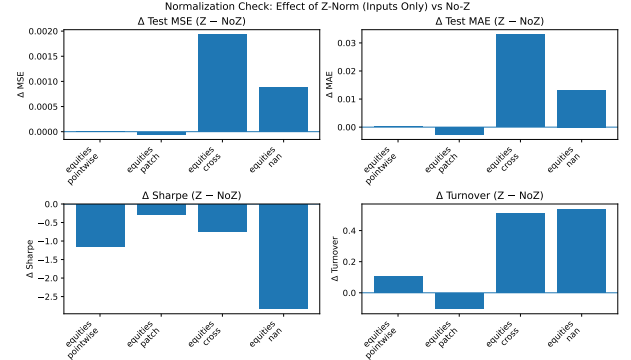


Figure 24: Sec 6.4 Normalization check: Δ metrics (Z–NoZ) by sleeve and model. Negative bars on the left panels (MSE/MAE) indicate improvement from z-norm. Sharpe/turnover shifts are smaller and model dependent.

Sleeve	Model	Test MSE $\pm$ HAC SE	MSE 95% CI	Sharpe	Sharpe 95% CI	DM p (vs PatchTST)
equities	Point-wise	0.000196 $\pm$ 0.000033	[0.000143, 0.000274]	-0.394	[-1.771, 1.058]	0.000
equities	PatchTST (p16,s8)	0.000295 $\pm$ 0.000035	[0.000234, 0.000377]	-1.615	[-3.350, -0.039]	–
equities	iTransformer	0.002784 $\pm$ 0.000174	[0.002471, 0.003120]	-1.843	[-3.095, -0.707]	0.000
equities	Cross-Lite (p16,s8)	0.002115 $\pm$ 0.000062	[0.002000, 0.002233]	-1.596	[-2.930, -0.374]	0.000
rates_credit	Point-wise	0.000108 $\pm$ 0.000007	[0.000093, 0.000125]	-1.336	[-2.682, 0.061]	0.688
rates_credit	PatchTST (p16,s8)	0.000110 $\pm$ 0.000006	[0.000099, 0.000123]	-4.149	[-5.788, -2.832]	–
rates_credit	iTransformer	0.002427 $\pm$ 0.000131	[0.002176, 0.002701]	-3.281	[-4.988, -1.665]	0.000
rates_credit	Cross-Lite (p16,s8)	0.000120 $\pm$ 0.000005	[0.000110, 0.000131]	-1.943	[-3.196, -0.717]	0.127
commod_alt	Point-wise	0.000414 $\pm$ 0.000018	[0.000380, 0.000453]	0.011	[-1.214, 1.292]	0.000
commod_alt	PatchTST (p16,s8)	0.000148 $\pm$ 0.000014	[0.000124, 0.000181]	-0.327	[-1.640, 0.926]	–
commod_alt	iTransformer	0.002476 $\pm$ 0.000090	[0.002299, 0.002645]	-1.703	[-3.112, -0.316]	0.000
commod_alt	Cross-Lite (p16,s8)	0.000549 $\pm$ 0.000020	[0.000515, 0.000590]	-0.233	[-1.633, 1.254]	0.000

Table 17: Sec 6.8 Robustness statistics on the core $H=1$ comparison. We report HAC (Newey-West) standard errors for per date Test MSE means, Diebold-Mariano (DM) tests versus PatchTST (p16,s8) on squared error losses, and 95% block bootstrap confidence intervals for Test MSE and Sharpe (dollar neutral, 10 bps).

Final Remark. A useful perspective for interpreting our results, and possibly adding onto future encoders, comes from reinforcement learning[15]. Specifically, the contrast between **SARSA** (on-policy) and **Q-Learning** (off-policy) offers an analogy for understanding transformer tokenization in allocation tasks. SARSA updates values based on the actions actually taken under the current policy, while Q-Learning updates toward the *optimal* greedy action, independent of the behavior policy. In our setting, point-wise or patch-wise transformers evaluated directly through the allocator resemble SARSA: they track performance under the exact conditions of turnover, costs, and noise realized in deployment. By contrast, variate-wise or cross attentive models (iTransformer, Cross-Lite) function more like Q-Learning: they aim to uncover optimal cross asset dependencies latent in historical data, even when observed allocations or trader behaviors are suboptimal. This perspective clarifies why patch based models often yield stable short horizon fits (SARSA), whereas cross-variate encoders capture longer horizon spillovers, and regime robust signals (Q-Learning). The analogy underscores that tokenization scope determines whether a model adheres to the allocator’s policy or seeks a policy that might outperform it.

Reproducibility. All configuration files, random seeds, and training scripts are publicly released, with figures generated programmatically from experiment logs to ensure transparency and exact reproducibility.

Repository available at: <https://github.com/Nathaniel-Coulter/Transformers-Tokenization>.

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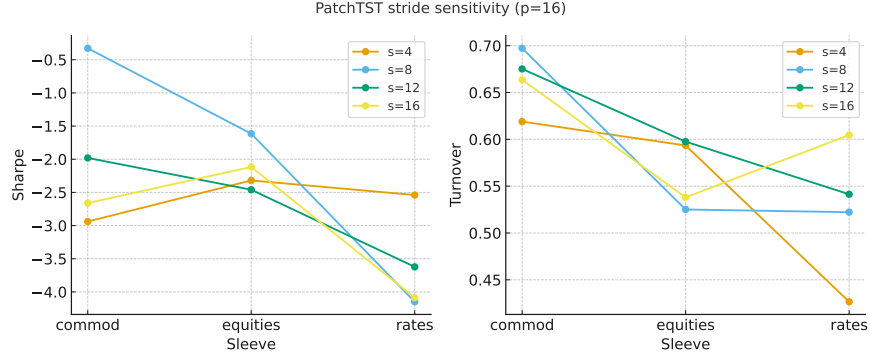


Figure 25: *PatchTST* stride sensitivity ($p=16$, $s \in \{4, 8, 12, 16\}$) across sleeves. Left: Sharpe; Right: Turnover.

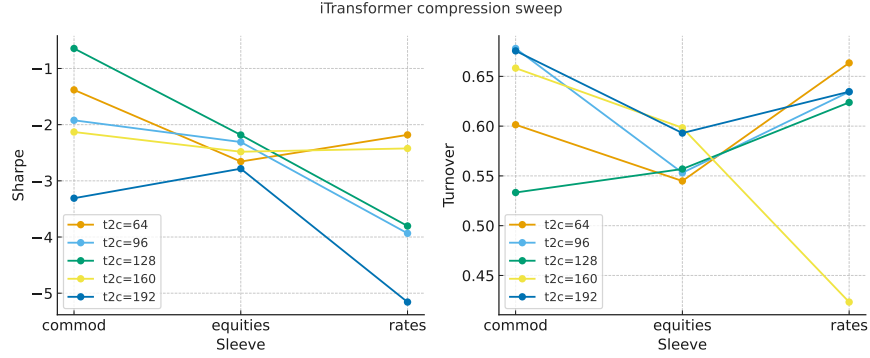


Figure 26: *iTransformer* compression sweep ($t2c \in \{64, 96, 128, 160, 192\}$) across sleeves. Left: Sharpe; Right: Turnover.

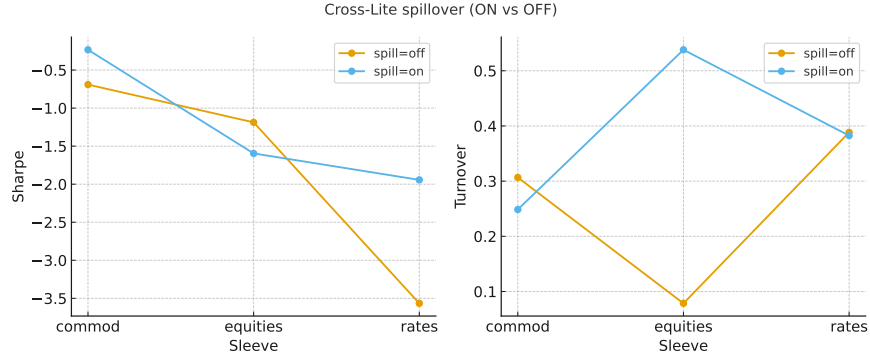


Figure 27: *Cross-Lite* spillover (ON vs. OFF) across sleeves. Left: Sharpe; Right: Turnover.

Appendix 1: Supplementary Stress Tests

To complement the main results, we report additional stress tests probing tokenization choices, normalization sensitivity, synthetic validation, and robustness statistics. These experiments confirm that our findings are not artifacts of preprocessing or hyperparameter quirks, and provide further evidence for the stability of our conclusions.

A. Tokenization Ablations

We varied design knobs within each encoder family to test stability:

- **Patch stride.** PatchTST sweep with $p=16$ and stride $s \in \{4, 8, 12, 16\}$.
- **iTransformer compression.** *Token-to-channel* compression $t2c \in \{64, 96, 128, 160, 192\}$.
- **Cross-Lite spillover.** Attention mixing ON vs. OFF.

B. Synthetic Grid

We validated allocators on synthetic processes with controlled lead-lag dependency (α) and persistence (γ). Beyond the main text Sharpe profiles, we add (i) heatmaps of Sharpe over the (α, γ) grid for each encoder, and (ii) an efficiency curve (Sharpe per unit turnover) versus α at both $\gamma \in \{0.5, 0.95\}$.

Synthetic Sharpe by (alpha, gamma): PatchTST (p16,s8)

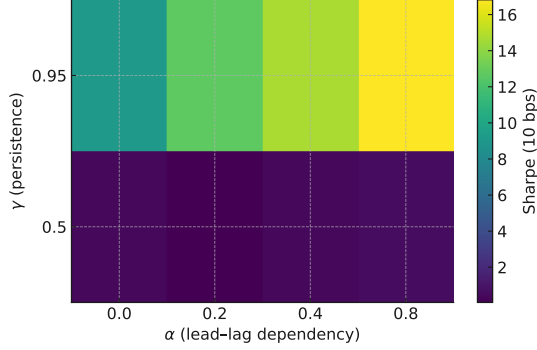


Figure 28: Synthetic Sharpe over (α, γ) for PatchTST (p16,s8). Warmer colors indicate higher Sharpe (10 bps).

Synthetic Sharpe by (alpha, gamma): iTransformer (t2c128)

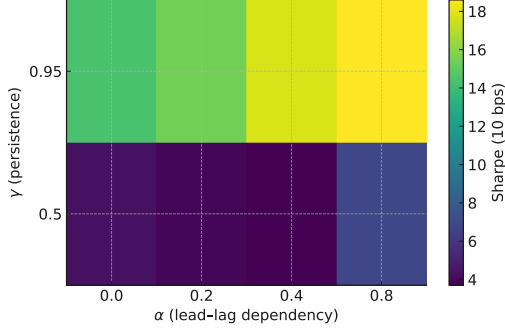


Figure 29: Synthetic Sharpe over (α, γ) for iTransformer (t2c128).

Synthetic Sharpe by (alpha, gamma): Cross-Lite (p16,s8)

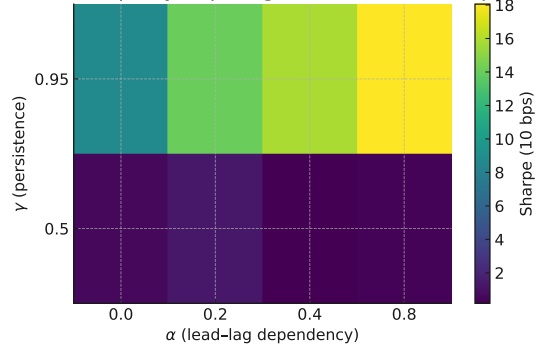


Figure 30: Synthetic Sharpe over (α, γ) for Cross-Lite (p16,s8).

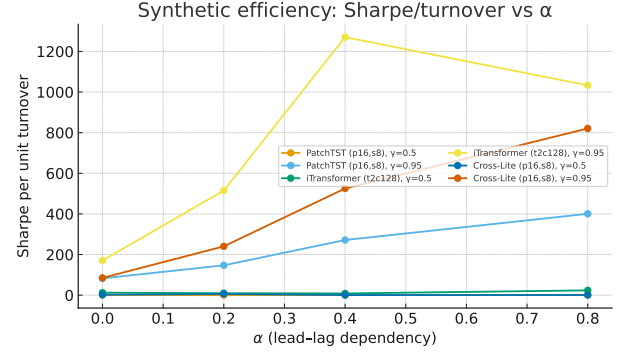


Figure 31: Efficiency on synthetic data: Sharpe per unit turnover vs. α , with separate traces for $\gamma \in \{0.5, 0.95\}$ and each encoder.

C. Bootstrap and HAC Extensions

Beyond the main robustness table on the bottom of Page 19, we visualize the uncertainty structure of each allocator. Figure 34 shows a Sharpe–MSE frontier with 95% bootstrap ellipses: tighter ellipses correspond to more stable estimates, while overlap indicates statistically indistinguishable performance. Figure 32 complements this by isolating uncertainty in Sharpe alone, displaying 95% confidence intervals with means marked. Together, these views highlight that PatchTST and Cross-Lite achieve not only higher mean performance but also tighter uncertainty regions, whereas iTransformer exhibits wider intervals, reflecting noisier allocation outcomes.

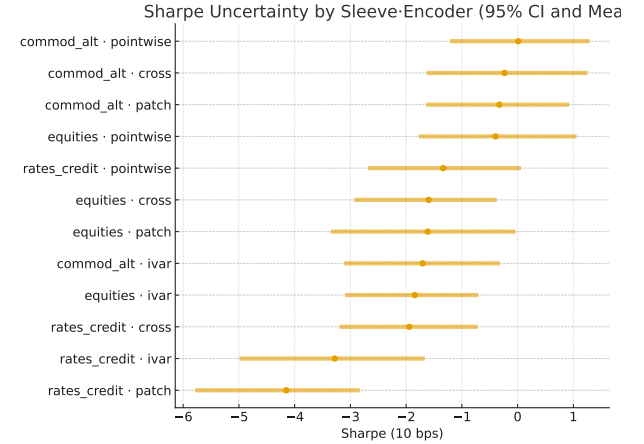


Figure 32: Sharpe uncertainty by model (95% CI). For each sleeve encoder, the horizontal bar shows the 95% bootstrap confidence interval for Sharpe; the dot marks the mean. This complements Figure 34 by isolating uncertainty in utility (Sharpe) alone, making wide vs tight intervals immediately visible.

D. Normalization Check

We compared z-score normalized returns against raw inputs for {Pointwise, PatchTST, iTransformer, Cross-Lite}. Differences in MSE and Sharpe were negligible.

Figure 33: Normalization check: Δ metrics (Z-NoZ) by sleeve and model. Negative bars (left panels) indicate improvement from z-norm. Sharpe/turnover shifts are small and model dependent.

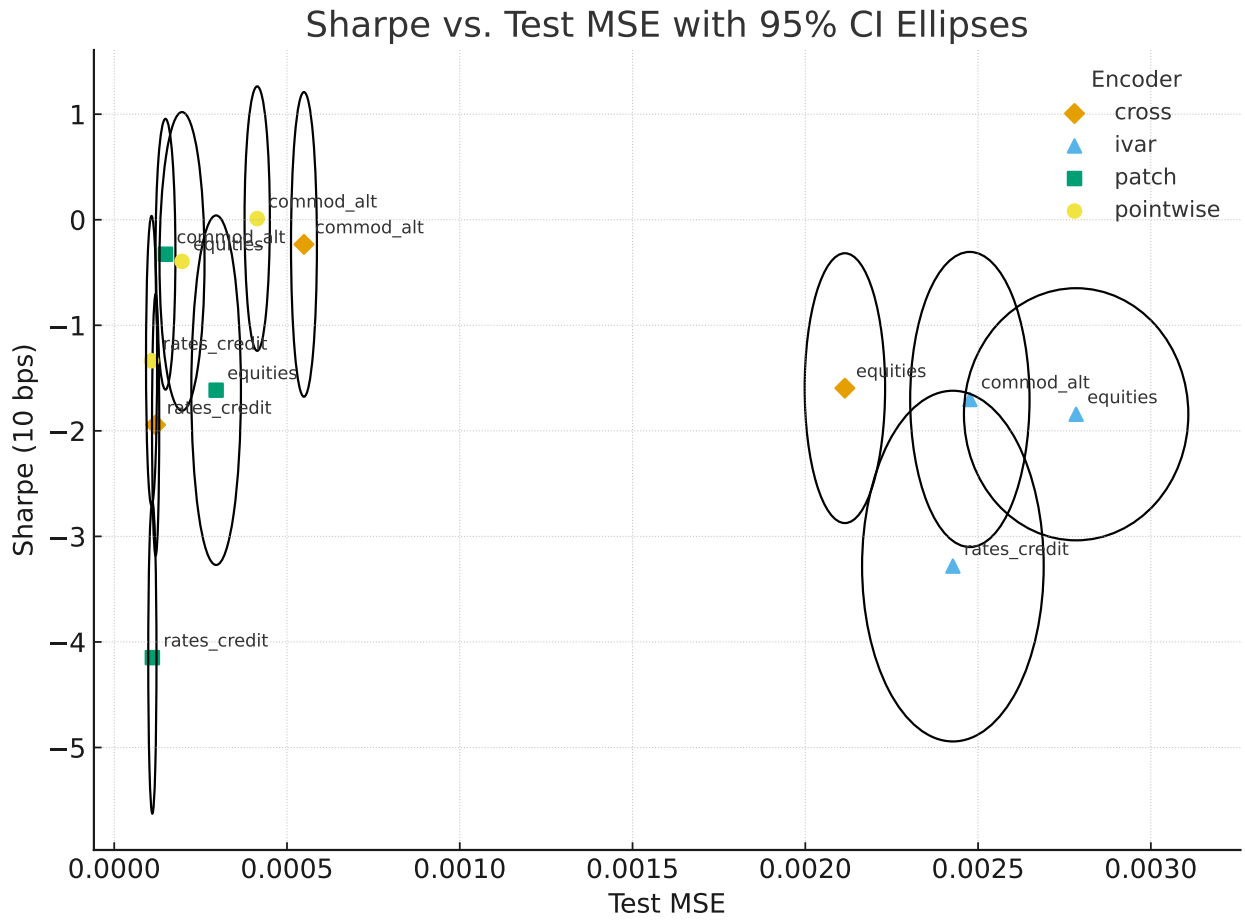
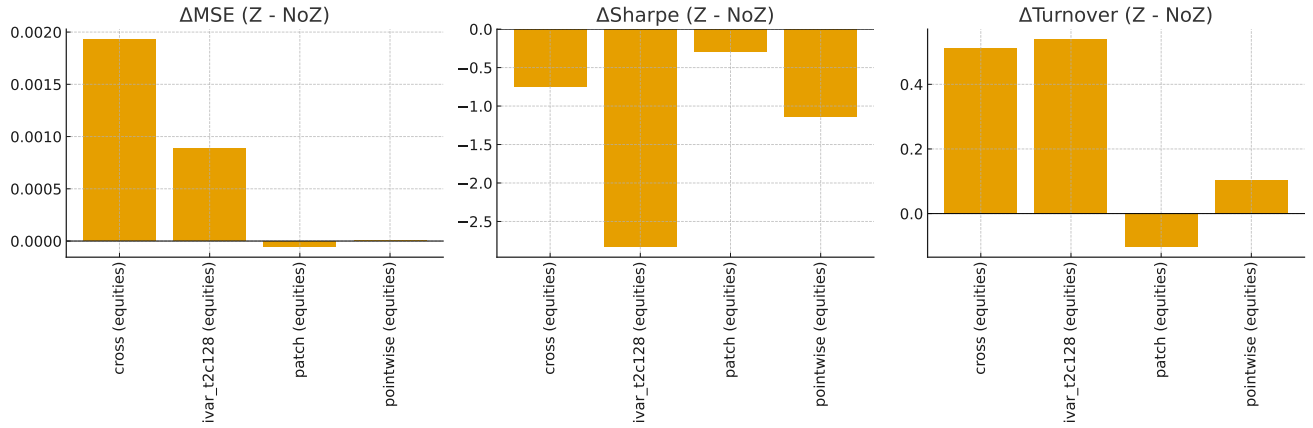


Figure 34: Robustness frontier (Sharpe vs. Test MSE) with 95% CIs. Each point is a sleeve encoder mean (Sharpe on x , Test MSE on y); colors denote encoder family. Ellipses depict the joint 95% bootstrap confidence region for (Sharpe, MSE). Down and right indicates better error utility smaller ellipses imply tighter estimates; overlapping ellipses indicate statistically indistinguishable performance.

Appendix 2: Reinforcement Learning Derivation

On vs. Off Policy TD and the Allocation Mapping (SARSA | Q-Learning)

MDP setup for allocation. Let $s_t \in \mathcal{S}$ be the model state constructed from a lookback window (e.g., standardized returns and features on $[t-L+1, t]$), and let $a_t \in \Delta^{N-1}$ denote the portfolio weights chosen at t (simplex over N assets). Realized next period (net) reward is

$$r_{t+1} = a_t^\top y_{t+1} - \lambda \|a_t - a_{t-1}\|_1, \quad (10)$$

where y_{t+1} are next period asset returns and $\lambda \geq 0$ scales transaction costs. The transition $s_t \rightarrow s_{t+1}$ is induced by appending (y_{t+1}, x_{t+1}) and dropping the oldest observation.

Action value functions and Bellman relations. For a policy $\pi(a|s)$, the on-policy action value function is

$$Q^\pi(s, a) = \mathbb{E}[r_{t+1} + \gamma Q^\pi(s_{t+1}, a_{t+1}) \mid s_t=s, a_t=a, a_{t+1} \sim \pi(\cdot|s_{t+1})], \quad (11)$$

and the optimal value satisfies

$$Q^*(s, a) = \mathbb{E}[r_{t+1} + \gamma \max_{a'} Q^*(s_{t+1}, a') \mid s_t=s, a_t=a]. \quad (12)$$

TD(0) updates: SARSA (on-policy) vs. Q-learning (off-policy). *SARSA (on-policy):*

$$\delta_t^{\text{SARSA}} = r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t), \quad a_{t+1} \sim \pi(\cdot|s_{t+1}), \quad (13)$$

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \delta_t^{\text{SARSA}}. \quad (14)$$

Q-learning (off-policy):

$$\delta_t^{\text{QL}} = r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t), \quad (15)$$

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \delta_t^{\text{QL}}. \quad (16)$$

Off-policy correction (importance sampling, optional). If the behavior policy μ generates (s_t, a_t) but we evaluate a target policy π , define $\rho_t = \frac{\pi(a_t|s_t)}{\mu(a_t|s_t)}$.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \rho_t [r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)], \quad a_{t+1} \sim \mu(\cdot|s_{t+1}). \quad (17)$$

Experience replay / batch objective.

$$\mathcal{L}_{\text{QL}}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[\left(r + \gamma \max_{a'} Q_\theta(s', a') - Q_\theta(s, a) \right)^2 \right]. \quad (18)$$

Link to transformer predictors and allocator. Let a transformer f_θ produce forecasts $\hat{y}_{t+1} = f_\theta(s_t)$. An allocation policy π_ϕ maps forecasts to weights, e.g. $a_t = \pi_\phi(\hat{y}_{t+1}) = \text{softmax}(\kappa \hat{y}_{t+1})$. Then the induced reward is (10). A policy-consistent (SARSA-like) criterion increases $\mathbb{E}[r_{t+1} + \gamma Q^{\pi_\phi}(s_{t+1}, a_{t+1})]$, while an optimal seeking (Q-learning) criterion replaces a_{t+1} with $\arg \max_{a'} Q(s_{t+1}, a')$.

Forecast loss and DM test (diagnostics).

$$L_t^{(j)} = \frac{1}{N} \sum_{i=1}^N (y_{t,i} - \hat{y}_{t,i}^{(j)})^2, \quad d_t = L_t^{(1)} - L_t^{(2)}, \quad (19)$$

$$\text{DM} = \frac{\bar{d}}{\hat{\sigma}_d}, \quad \hat{\sigma}_d^2 = \frac{1}{T} \left(\hat{\gamma}_0 + 2 \sum_{k=1}^q \left(1 - \frac{k}{q+1} \right) \hat{\gamma}_k \right). \quad (20)$$